EXPOSE PFSENSE



PREMIERE PARTIE:



PARTICIPANT : DARIUS, THADDE, CHRIST, EMMANUEL, CLAUDE



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- VPN (OpenVPN, IPsec, WireGuard)
- Sécurité avancée (IDS/IPS, pfBlockerNG)
- Supervision et diagnostic
- Conclusion

INTRODUCTION



- **PfSense:** *est une distribution de pare-feu et routeur libre basée sur FreeBSD. Elle est spécialement conçue pour fournir des fonctionnalités avancées de sécurité réseau, de routage, et de gestion de trafic, tout en restant accessible à travers une interface web riche et intuitive.*
- Créée par Netgate, pfSense *est utilisé aussi bien par les particuliers, les entreprises, les établissements scolaires que les fournisseurs de services.*

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OBJECTIF DE PFSENSE



pfSense a été conçu pour :

offrir une solution de pare-feu et de routage complète ;

être hautement configurable et extensible ;

fournir une alternative open source aux solutions professionnelles coûteuses (Fortinet, Cisco ASA, etc.)

permettre une administration facile via interface web ;

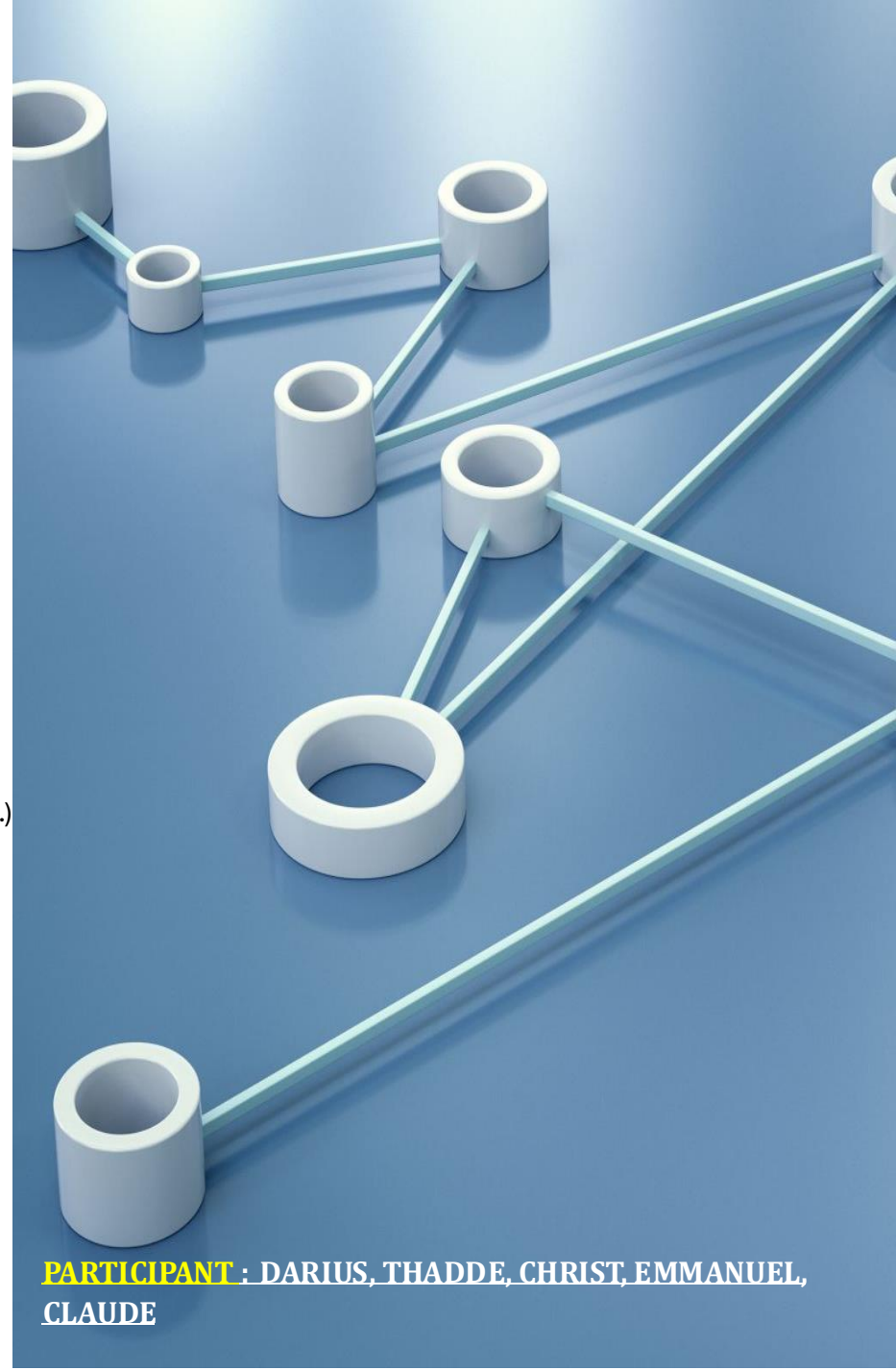
assurer une stabilité et sécurité proches de celles des systèmes professionnels.

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RÔLE DE PFSENSE

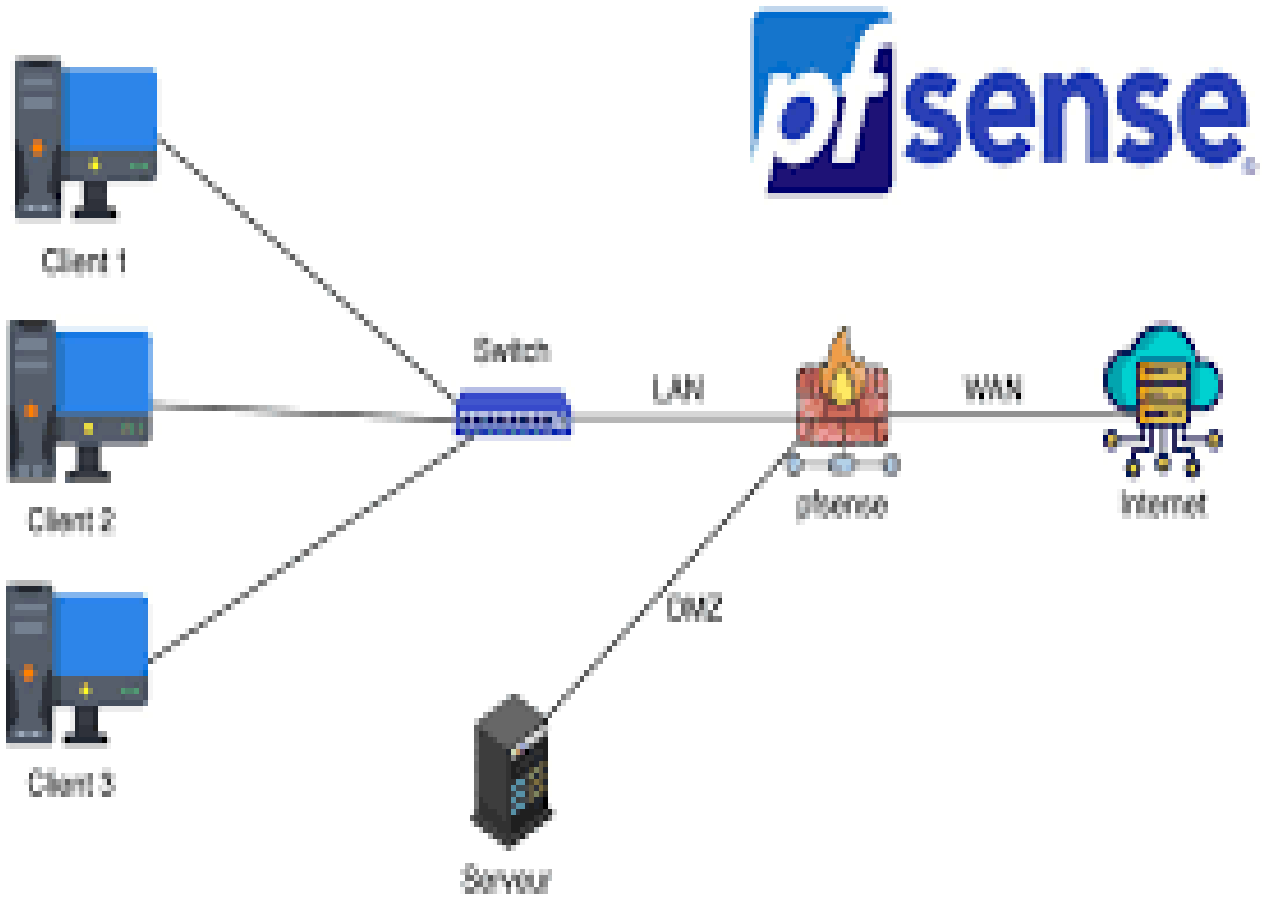
pfSense joue plusieurs rôles essentiels dans l'infrastructure réseau :

- **Pare-feu**
  - Filtrage des paquets entrants/sortants
  - Règles avancées par interface, protocole, port, utilisateur
  - Stateful Packet Inspection (SPI)
- **Routeur**
  - Gestion des routes statiques et dynamiques (via packages)
  - Support IPv4 et IPv6
  - Multi-WAN pour redondance et équilibrage de charge
- **Serveur VPN**
  - Il prend en charge différents protocoles VPN :
  - OpenVPN
  - IPsec
  - WireGuard (via plugin)
  - L2TP
- **Système de gestion de trafic**
  - QoS (priorisation du trafic)
  - Traffic shaping
  - Limitation de bande passante
- **Serveur DHCP / DNS**
  - DHCP complet avec réservation d'adresses
  - DNS Resolver / DNS Forwarder
  - Fonctionnalités DNS avancées (DNS over TLS, listes noires…)
- **Portail captif**
  - Authentification des utilisateurs réseau
  - Comptabilité
  - Redirection vers page personnalisée
- **Sécurité avancée**
  - IDS/IPS : Snort, Suricata
  - Filtrage web (avec Squid + SquidGuard)
  - Filtrage des menaces en temps réel



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ARCHITECTURE TECHNIQUE



4.1 Système d'exploitation

Basé sur FreeBSD

Fiabilité et performances élevées

Support matériel large (x86_64, ARM pour certains modèles Netgate)

4.2 Interface Web (GUI)

Accessible via navigateur

Organisation claire : Firewall, Services, System, Diagnostics, etc.

Sauvegarde/restauration de configuration en XML

4.3 Packages supplémentaires

pfSense se distingue par un large écosystème de modules :

Squid : proxy web

SquidGuard : filtrage de contenu

Snort / Suricata : IDS/IPS

pfBlockerNG : blocage avancé de pays, IP, pubs…

ACME : certificats Let's Encrypt

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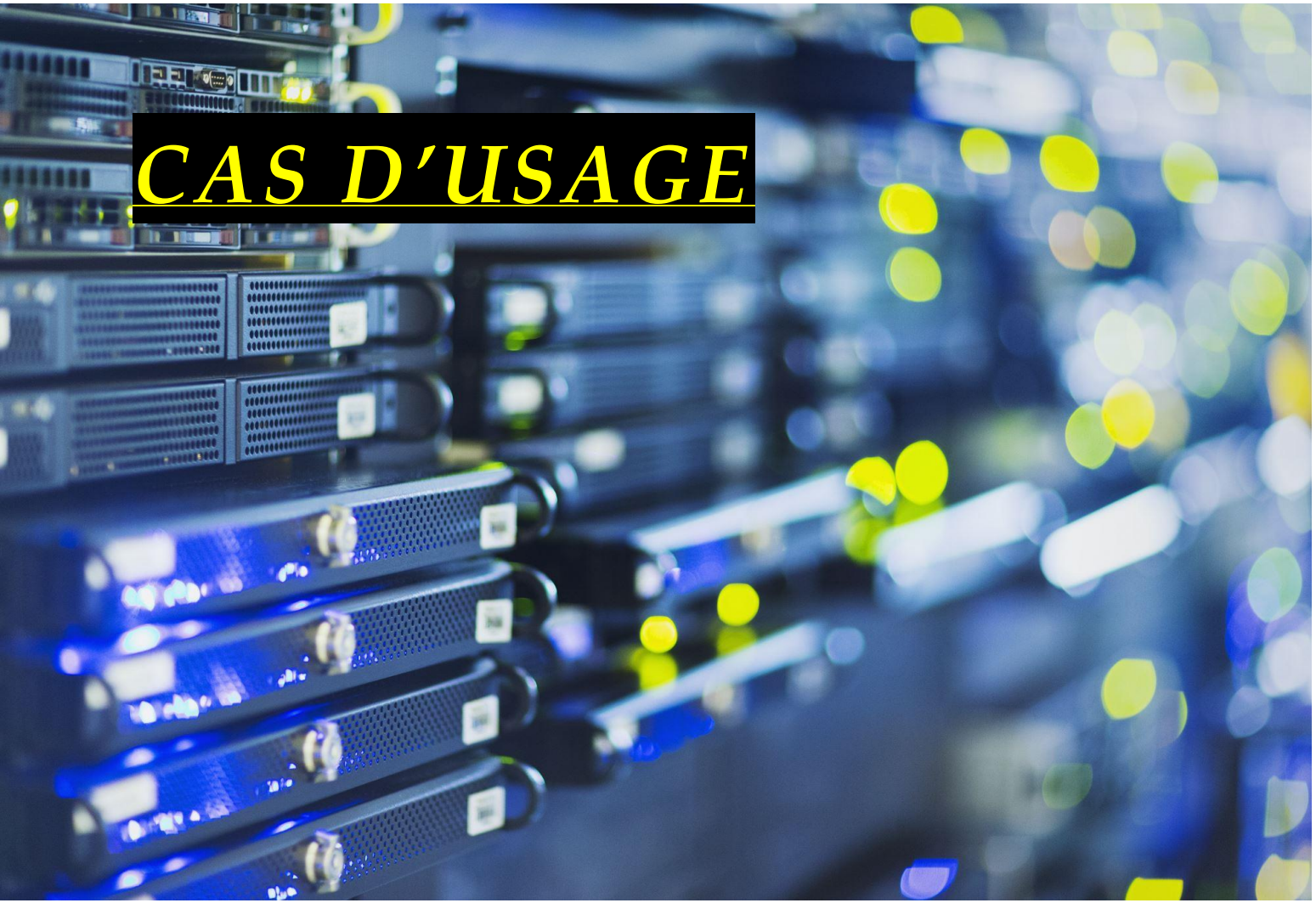
FONCTIONNALITÉS PRINCIPALES DÉTAILLÉES

- **5.1 Firewall**
- Moteur pf (Packet Filter)
- Règles configurables par interface LAN/WAN/DMZ
- NAT automatique ou manuel
- Aliases pour faciliter la gestion des IP/ports
- **5.2 Réseau et routage**
- VLAN 802.1Q
- Haute disponibilité (CARP)
- Multi-WAN : Failover, Load balancing
- PPPoE, DHCP, statique, etc.

- **5.3 VPN**
- OpenVPN
- Authentification locale ou LDAP/Radius
- Export facile avec package "client-export"
- IPsec
- Interconnexion de sites (site-to-site)
- Compatible avec Cisco, Mikrotik, etc.
- WireGuard
- VPN léger et performant
- **5.4 Monitoring et diagnostic**
- Graphiques de bande passante en temps réel
- Logs détaillés
- Outils intégrés : ping, traceroute, packet capture
- Tableau de bord entièrement personnalisable



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CAS D'USAGE

Réseau d'entreprise

Segmentation LAN/VLAN

VPN pour télétravail

Haute disponibilité

Établissements scolaires

Filtrage web + portail captif

Gestion d'un grand nombre d'utilisateurs

Particuliers avancés

HomeLab, serveurs, domotique

Bloquage publicités avec pfBlockerNG

Fournisseurs d'accès / hébergeurs

Routage avancé

Multi-WAN

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AVANTAGES DE PFSENSE



- Gratuit (version CE) et open source
- Interface web puissante
- Très modulable
- Communauté active
- Alternative crédible aux firewall professionnels
- Stabilité et sécurité héritées de FreeBSD

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LIMITES DE PFSENSE



COURBE D'APPRENTISSAGE POUR LES DÉBUTANTS



CERTAINES FONCTIONNALITÉS AVANCÉES DEMANDENT DES CONNAISSANCES RÉSEAU



LE FILTRAGE WEB OPTIMAL NÉCESSITE DES PACKAGES TIERS

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CONCLUSION

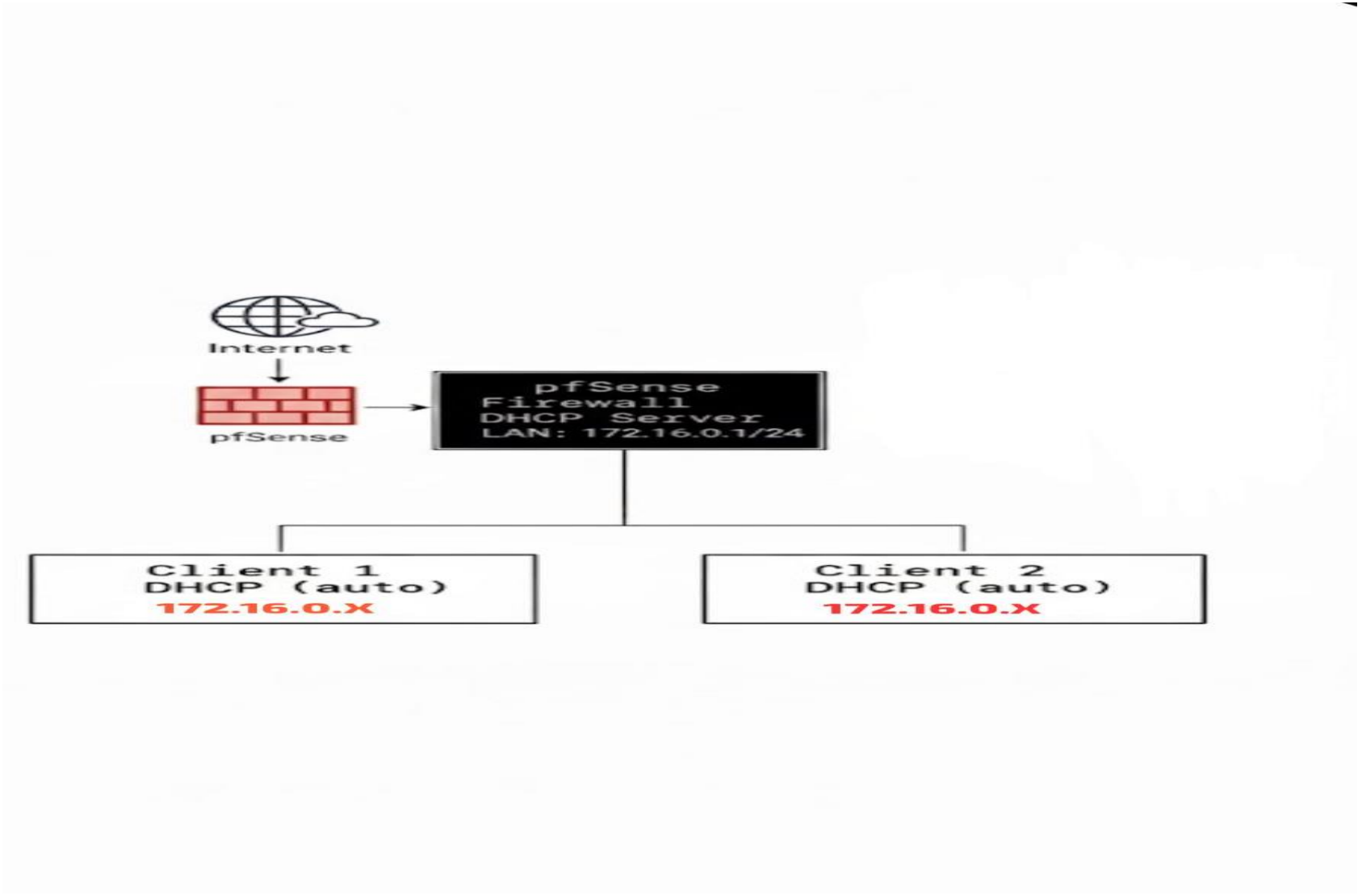


- pfSense : est une solution complète, fiable et extrêmement puissante pour la gestion de la sécurité et du routage réseau. Grâce à ses nombreuses fonctionnalités, son extensibilité via des packages et sa robustesse, il s'impose comme une référence incontournable dans le monde du firewall open source, aussi bien pour les particuliers technophiles que pour les entreprises.



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INFRASTRUCTURE RÉSEAU



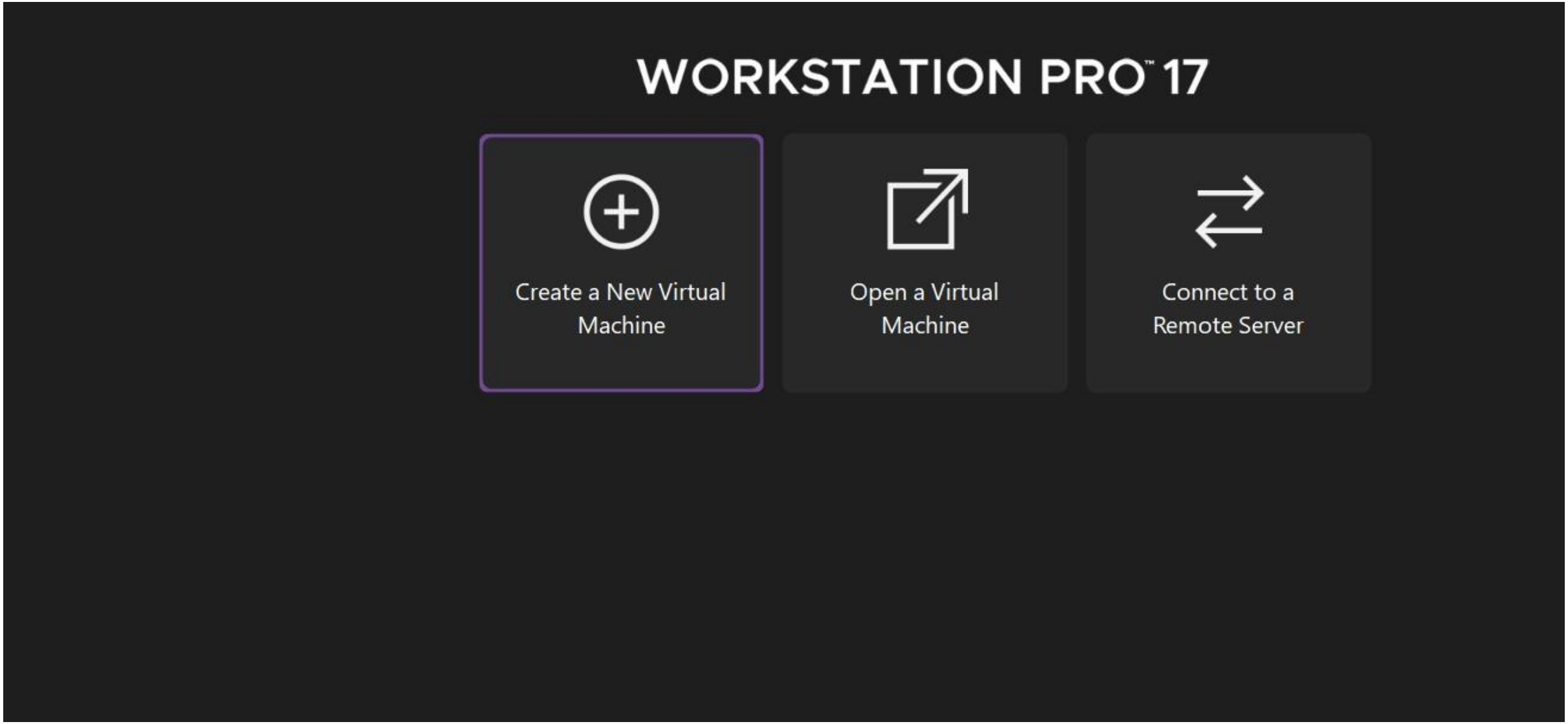
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DEUXIEME PARTIE:



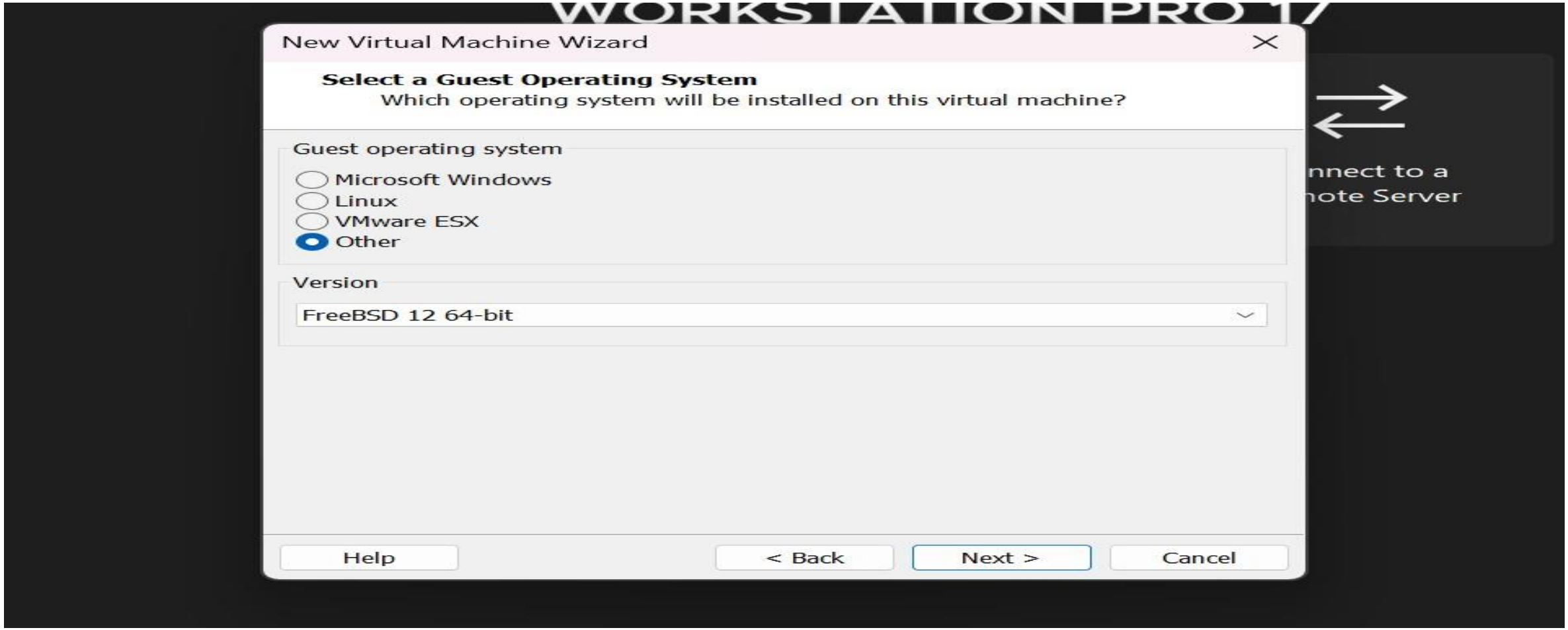
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INSTALLATION DE LA VM



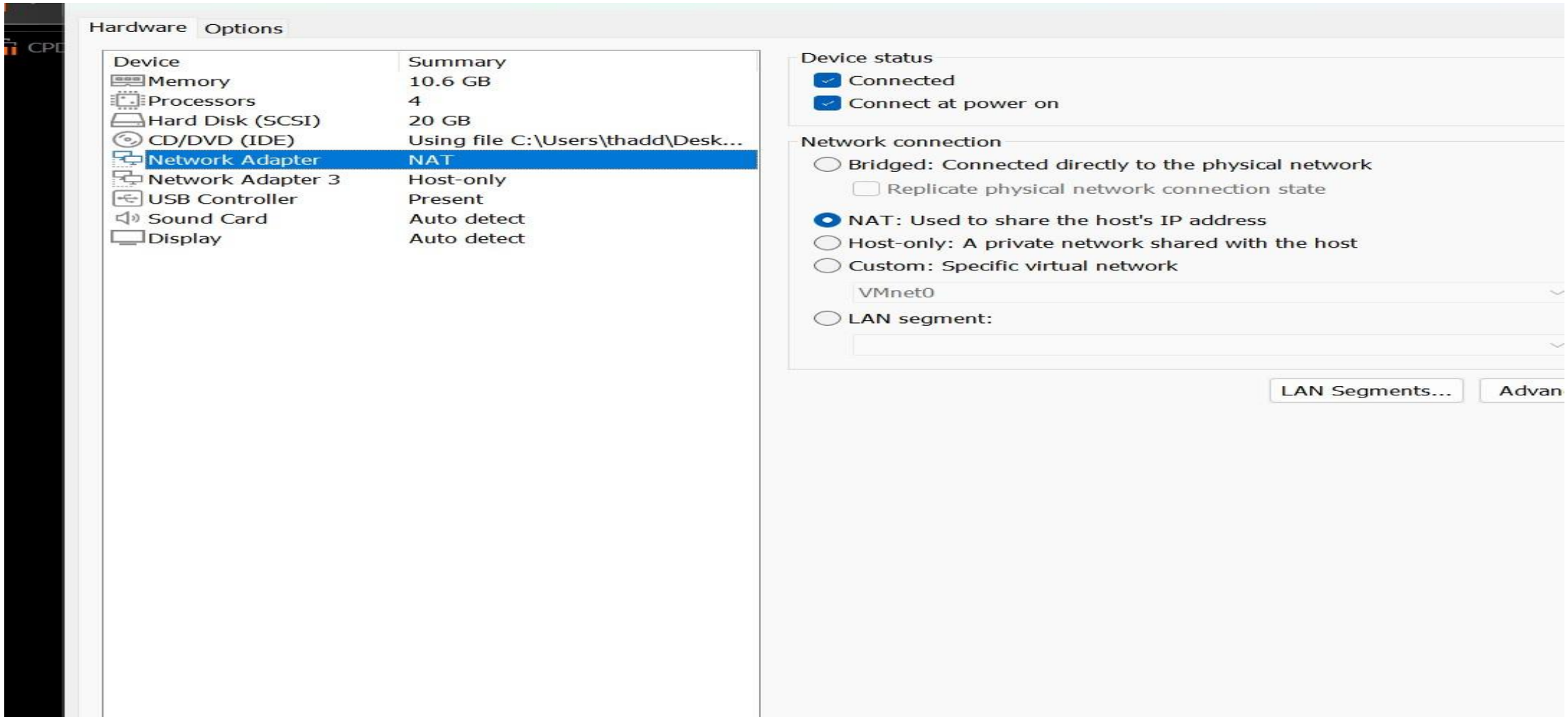
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CHOIX DU TYPE DE MACHINE ET DE LA VERSION



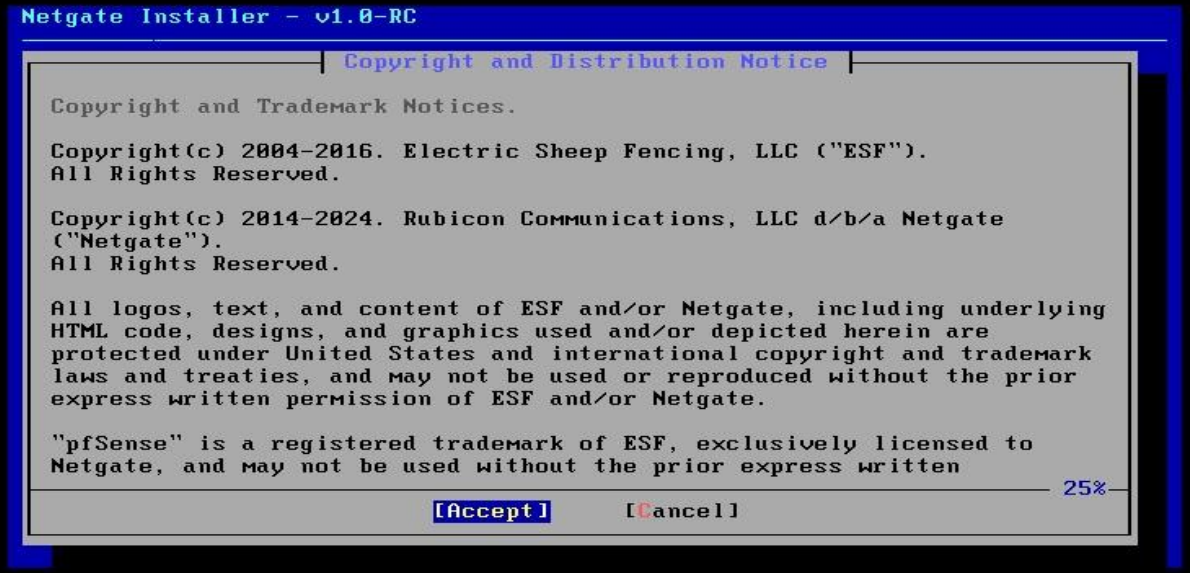
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CHOIX DU TYPE DE MACHINE ET DE LA VERSION



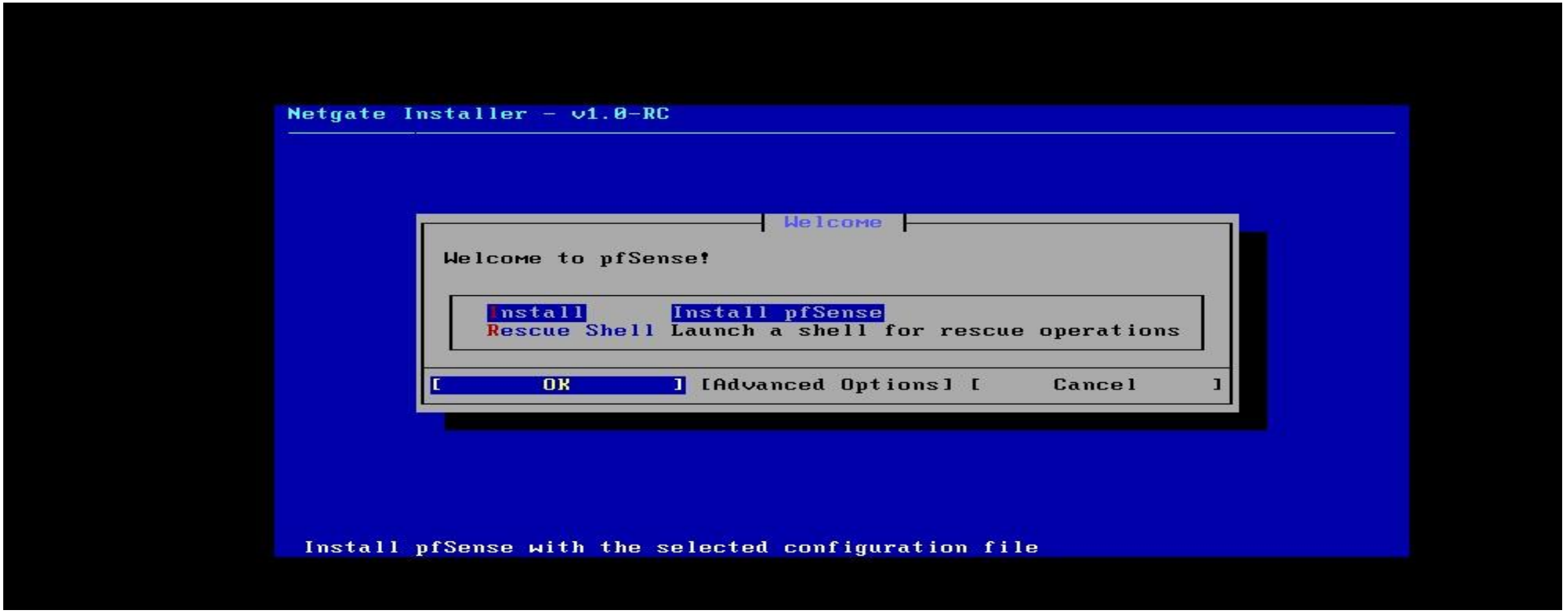
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ON CLIQUE SUR INSTALLER PFSENSE



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OUI C'EST LA MÊME CHOSE



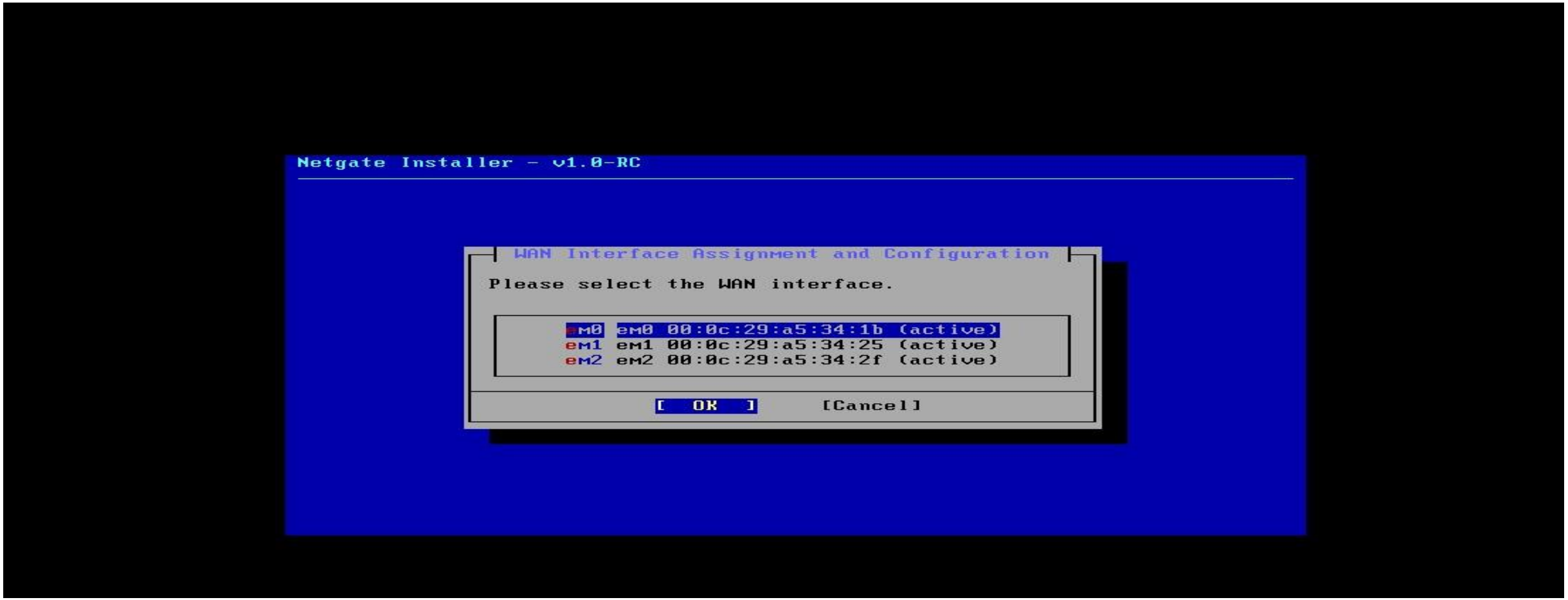
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ON CLIQUE SUR OK



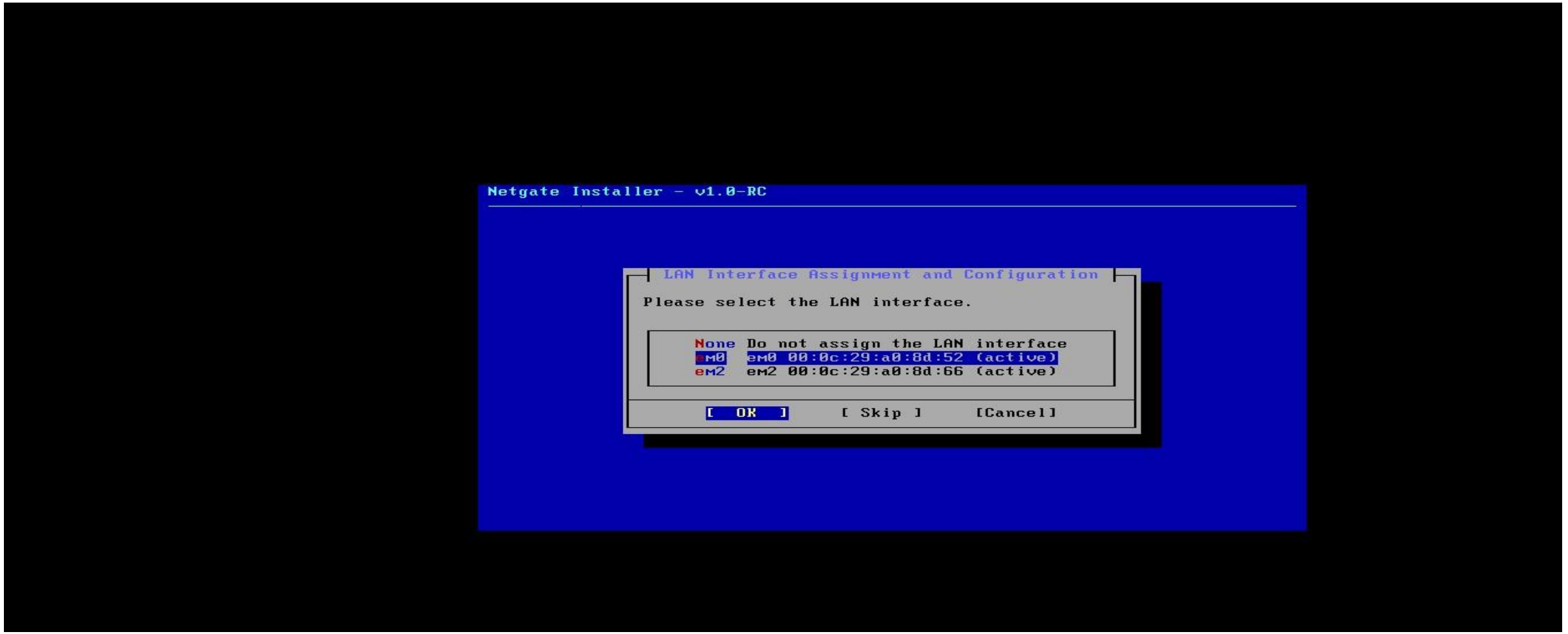
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ON CLIQUE SUR OK



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ON CHOISIT L'INTERFACE LAN



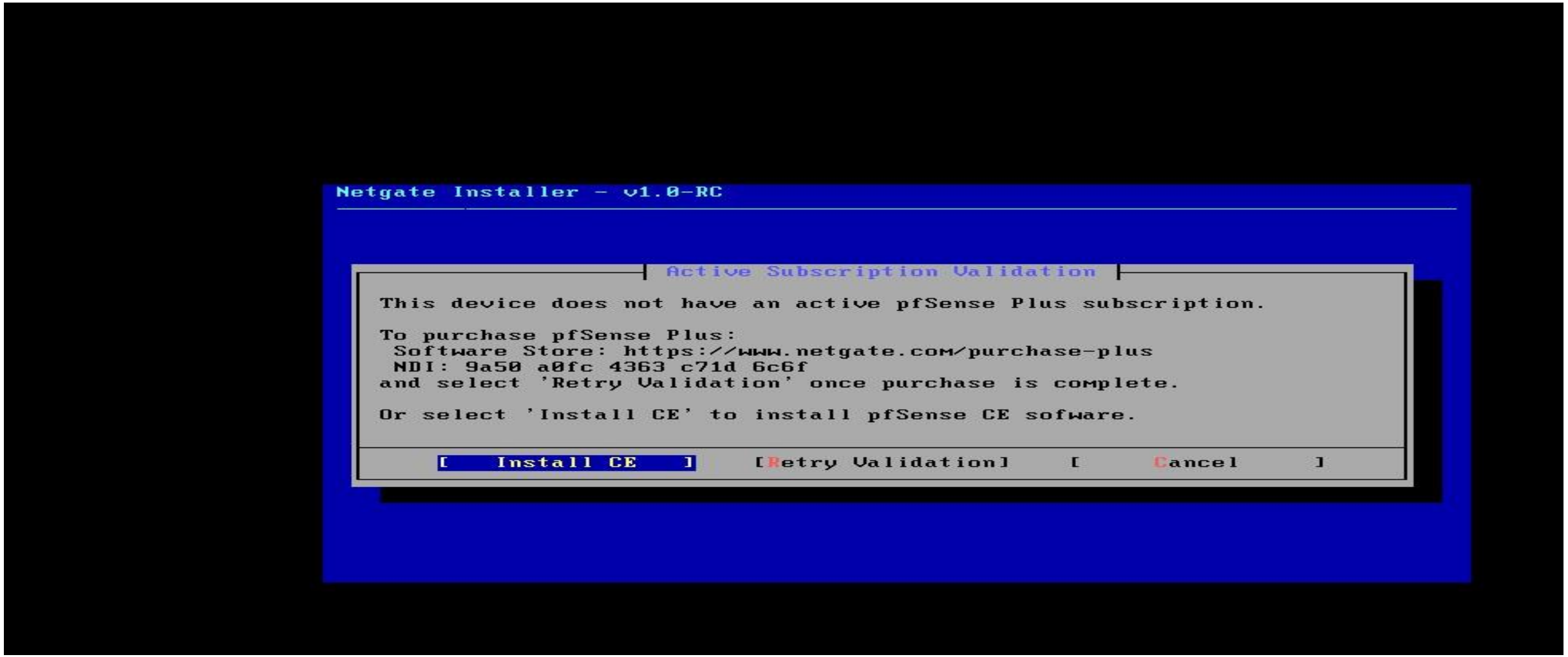
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ON CHOISIT LE TYPE DE PARTITION



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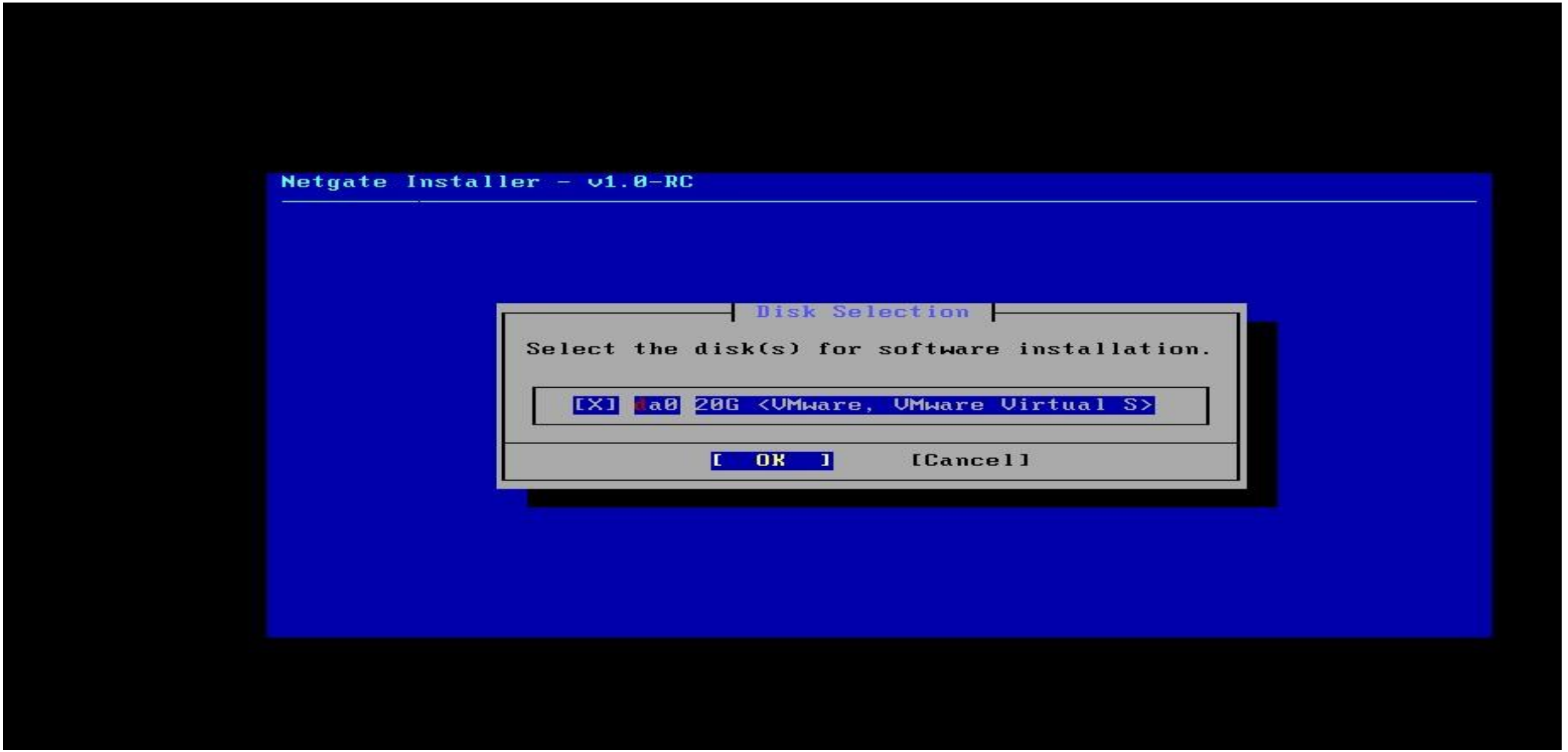
INSTALLATION DE PFSENSE EN COURS



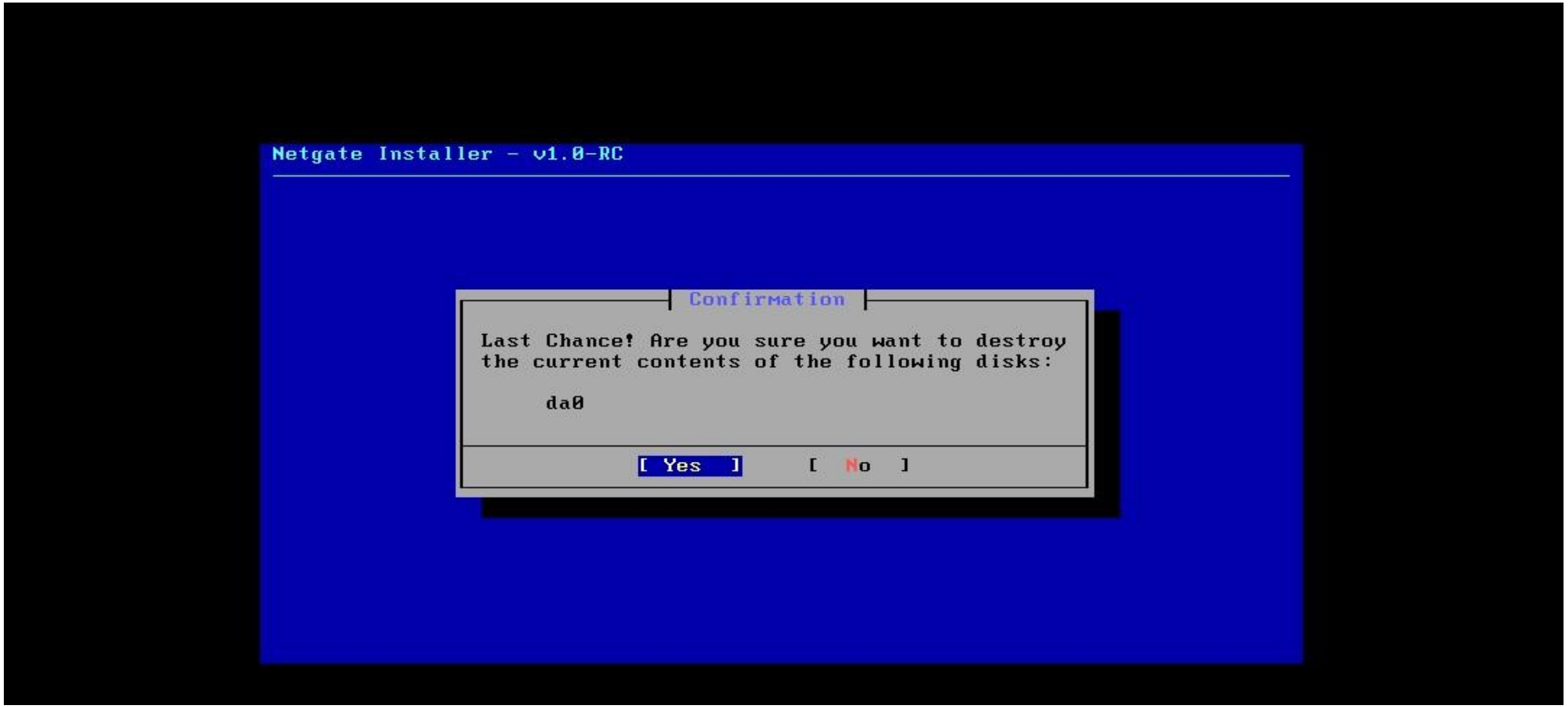
L'INSTALLATION EST FINI , ON CLIQUE SUR OK



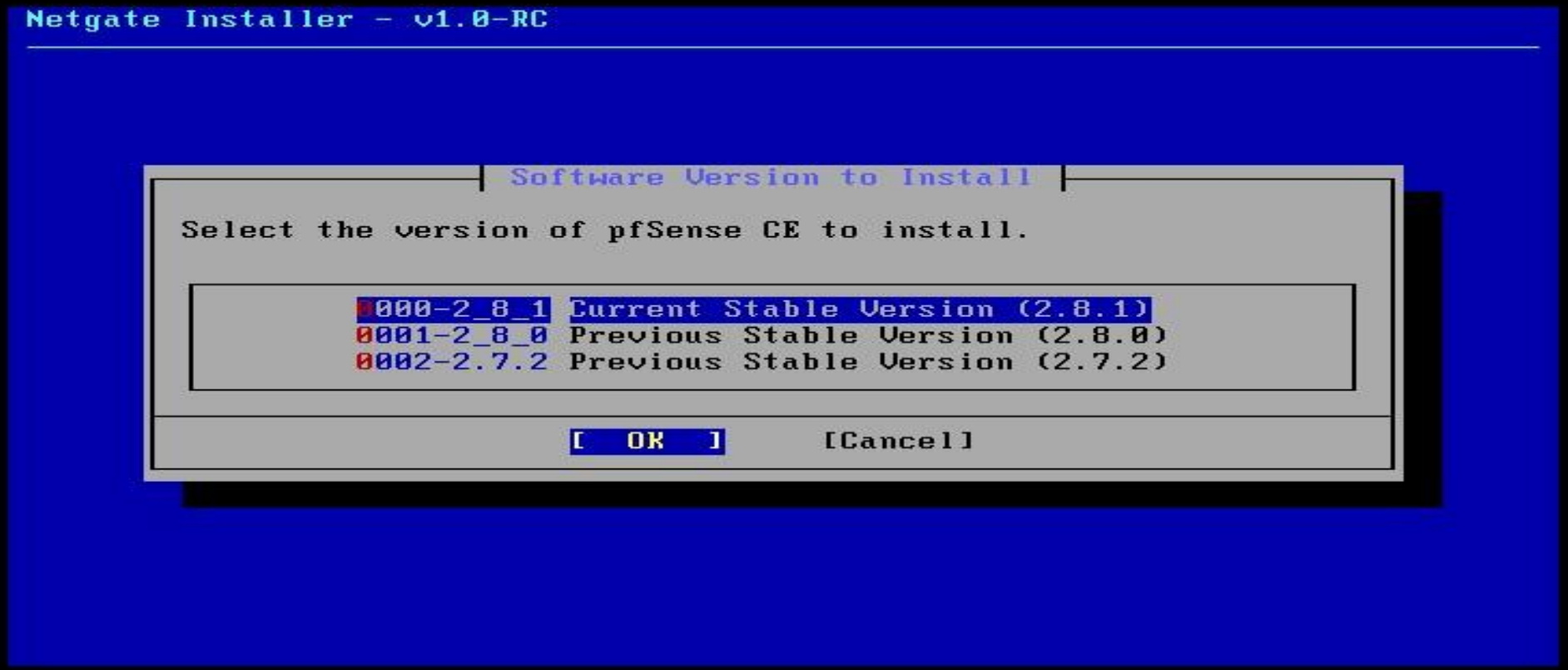
OK


Netgate Installer - v1.0-RC
Disk Selection
Select the disk(s) for software installation.
[X] da0 20G <VMware, VMware Virtual S>
[OK]
[Cancel]

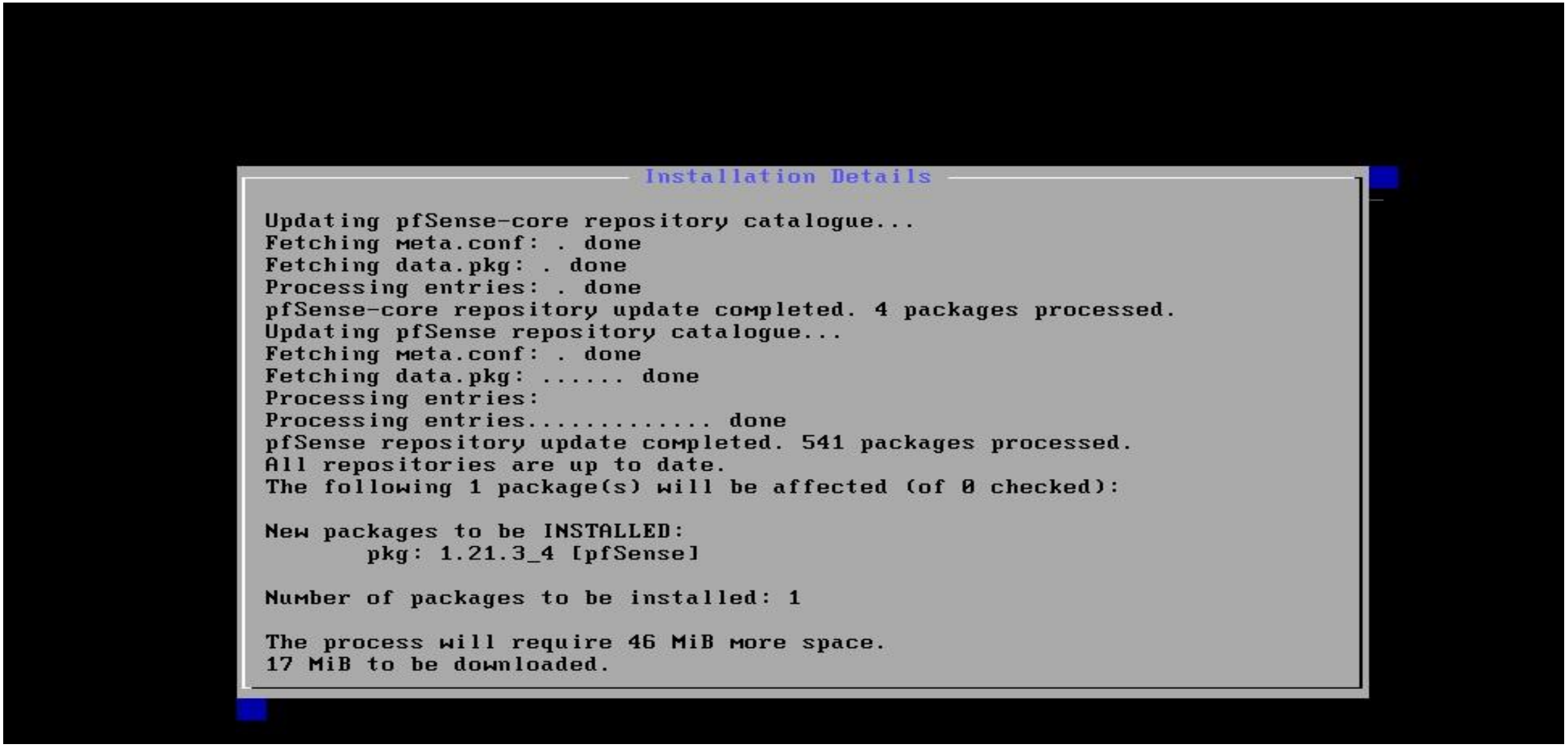
YES


Netgate Installer - v1.0-RC
Confirmation
Last Chance! Are you sure you want to destroy
the current contents of the following disks:
da0
[Yes]
[No]

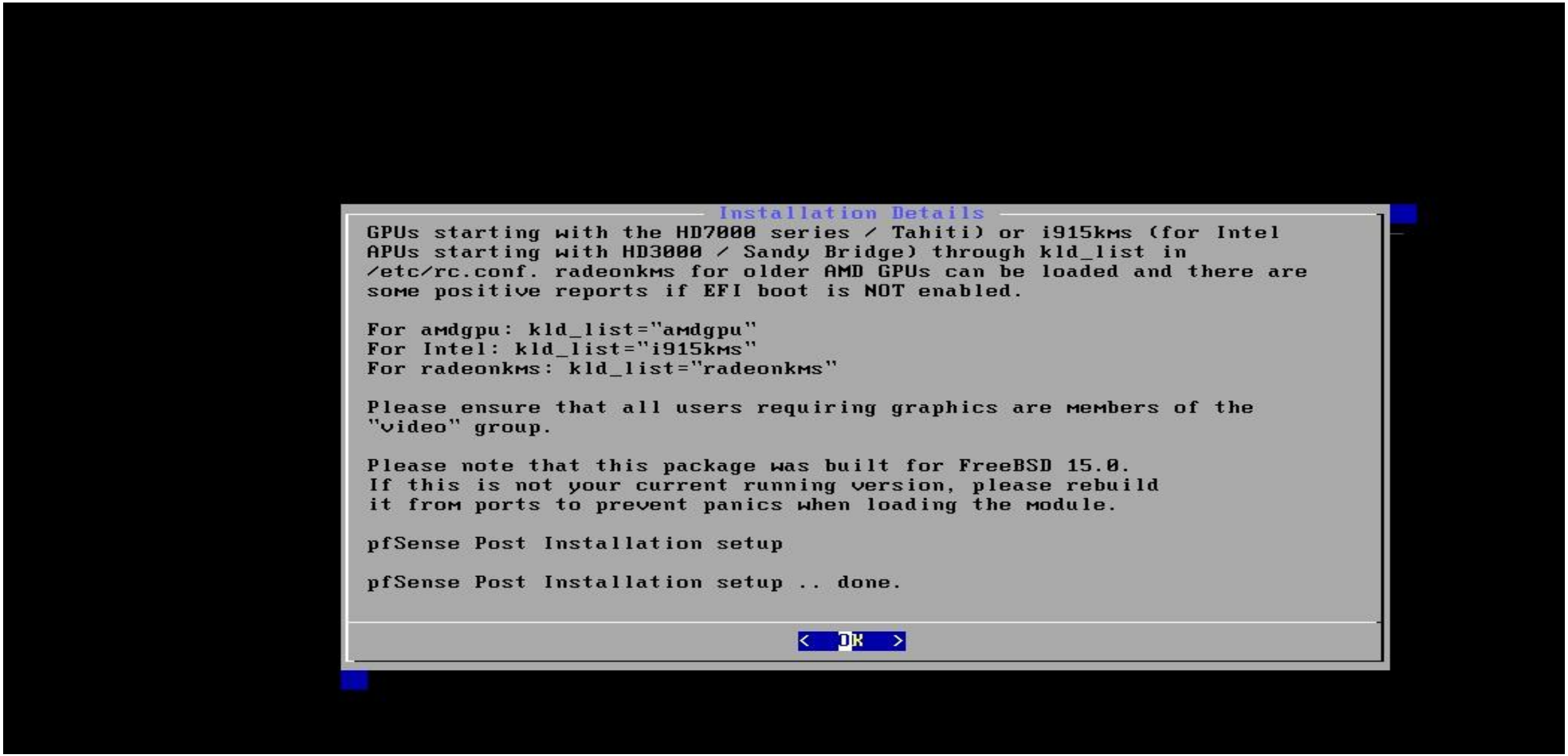
ON CHOISIT LE TYPE DE PARTITION



INSTALLATION DE PFSENSE EN COURS



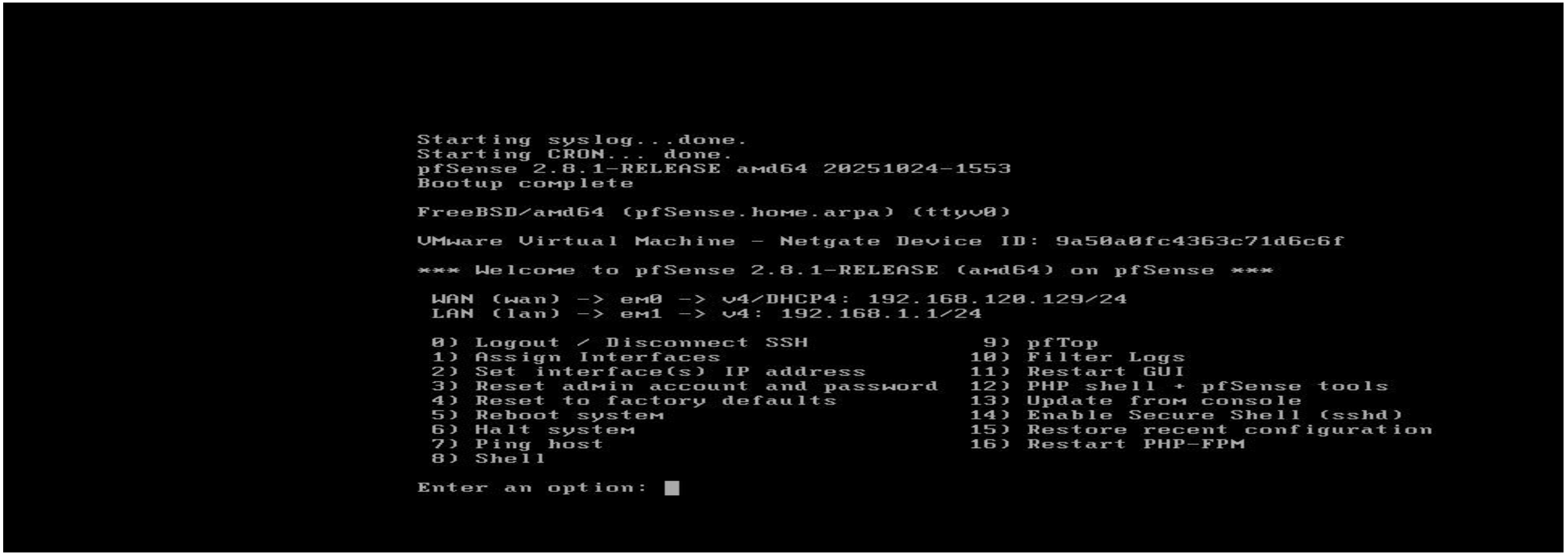
L'INSTALLATION EST FINI , ON CLIQUE SUR OK



ON AS CETTE PAGE , QUI APPARAÎT AVEC LES INTERFACES WAN ET LAN DONNER AUTOMATIQUEMENT GRÂCE AU DHCP DE MON ROUTEUR, ON LE CHANGERA PLUS TARD



ON TAPE 8 , POUR AVOIR ACCÈS AU POWER SHELL ET AINSI POUVOIR CHANGER LE CLAVIER EN AZERTY CAR DE BASE IL EST EN QWERTY



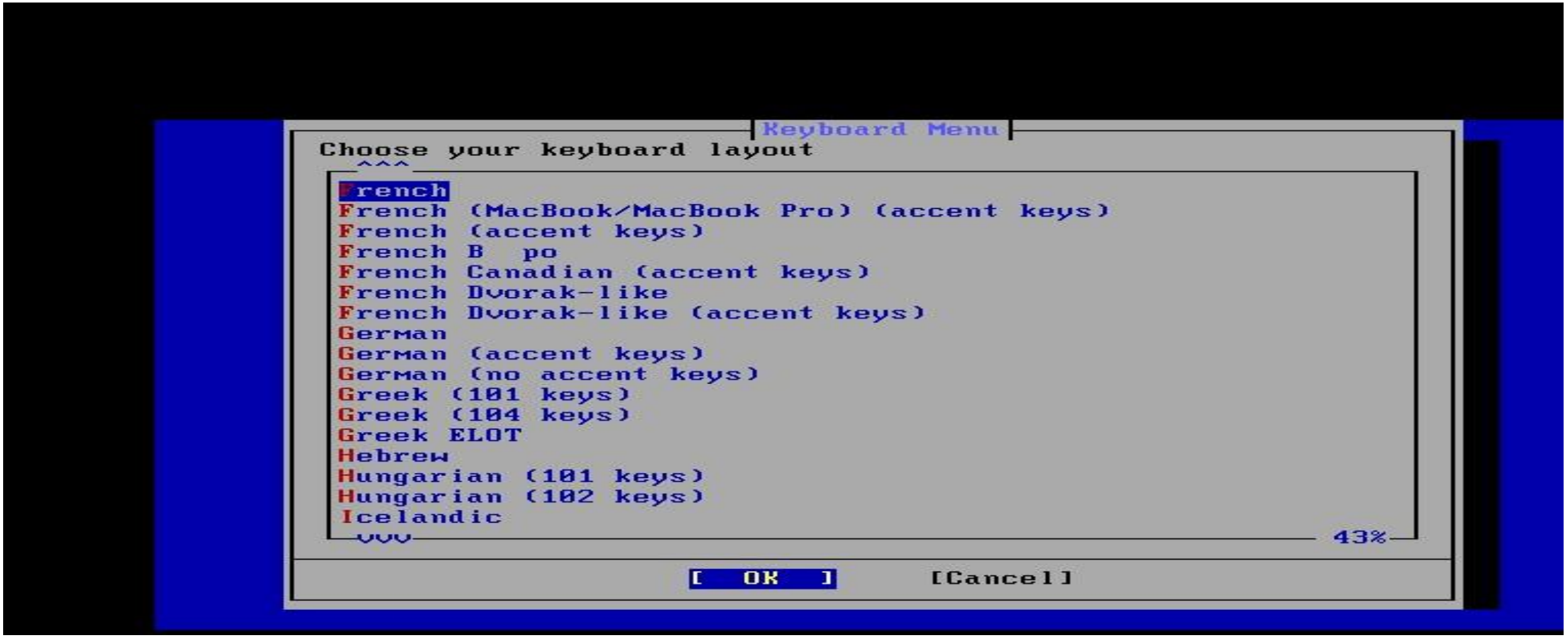
ON TAPE LA COMMANDE KBDMAP, POUR QUE APPARAISSENT LA FENÊTRE QUI VA NOUS PERMETTRE DE CHANGER DE CLAVIER



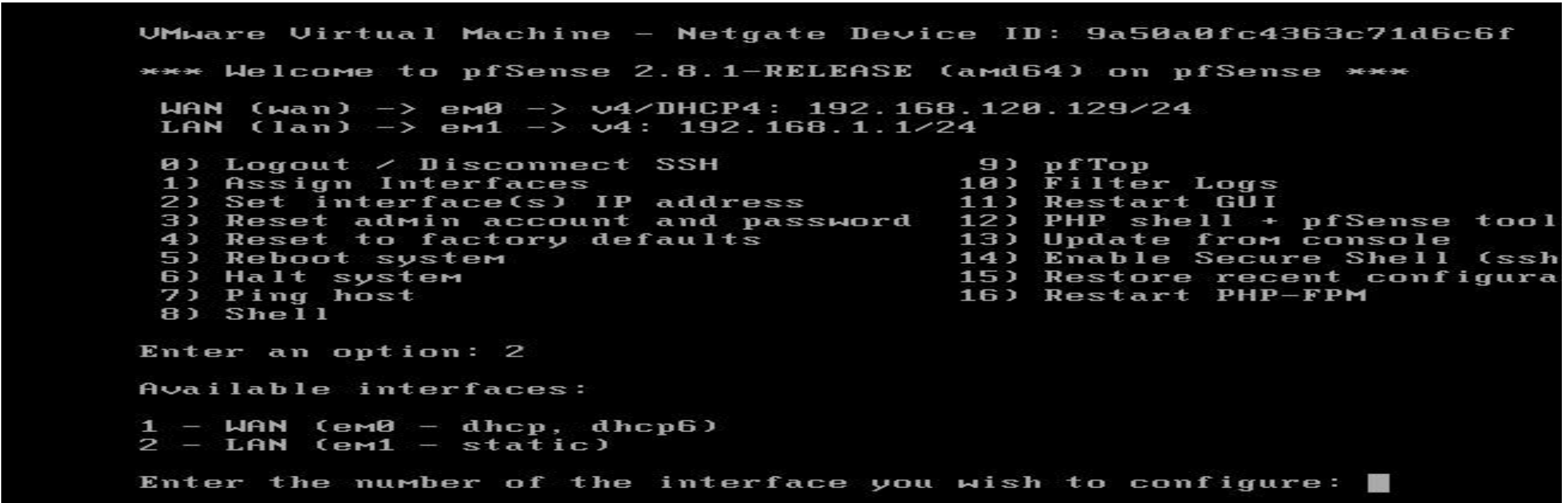
KBDMAP


pfSense 2.8.1-RELEASE amd64 20251024-1553
Bootup complete
FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)
VMware Virtual Machine - Netgate Device ID: 9a50a0fc4363c71
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 192.168.120.129/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSens
13) Update from conso
14) Enable Secure She
15) Restore recent co
16) Restart PHP-FPM
Enter an option: 8
[2.8.1-RELEASE][root@pfSense.home.arpa]/root: kbdmap

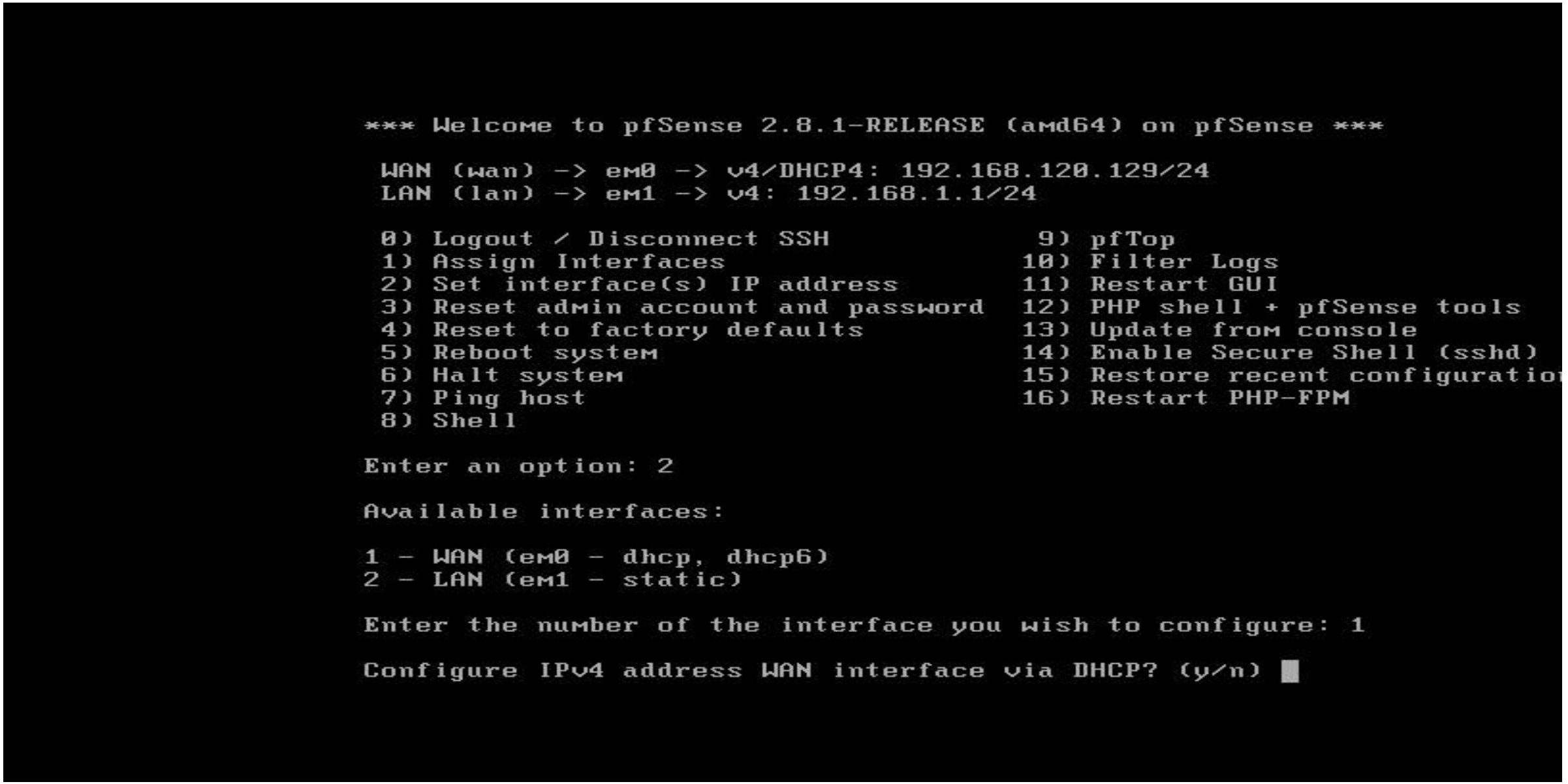
ON CHOISIT LE FRANÇAIS POUR AVOIR LE CLAVIER AZERTY



ON APPUIE SUR 2 , POUR CHANGER LES ADRESSES IP DE LAN ET WAN QUI ONT ÉTÉ ADRESSÉES AVEC LE DHCP DE MON ROUTEUR ON VA LES METTRE EN FIXE ET ADRESSE DES ADRESSE IP



YES


*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 192.168.120.129/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24
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6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuratio
16) Restart PHP-FPM
Enter an option: 2
Available interfaces:
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)
Enter the number of the interface you wish to configure: 1
Configure IPv4 address WAN interface via DHCP? (y/n)

YES


0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
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9) pfTop
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11) Restart GUI
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13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuratio
16) Restart PHP-FPM
Enter an option: 2
Available interfaces:
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)
Enter the number of the interface you wish to configure: 1
Configure IPv4 address WAN interface via DHCP? (y/n) n
Enter the new WAN IPv4 address. Press <ENTER> for none:
>
Configure IPv6 address WAN interface via DHCP6? (y/n) n

IP


LAN (lan) -> em1 -> v4: 192.168.1.1/24
0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM
Enter an option: 2
Available interfaces:
1 - WAN (em0)
2 - LAN (em1 - static)
Enter the number of the interface you wish to configure: 1
Configure IPv4 address WAN interface via DHCP? (y/n) n
Enter the new WAN IPv4 address. Press <ENTER> for none:
> 172.16.0.25

IP

```
 6) Halt system                                  15) Restore recent configuration
 7) Ping host                                    16) Restart PHP-FPM
 8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 1

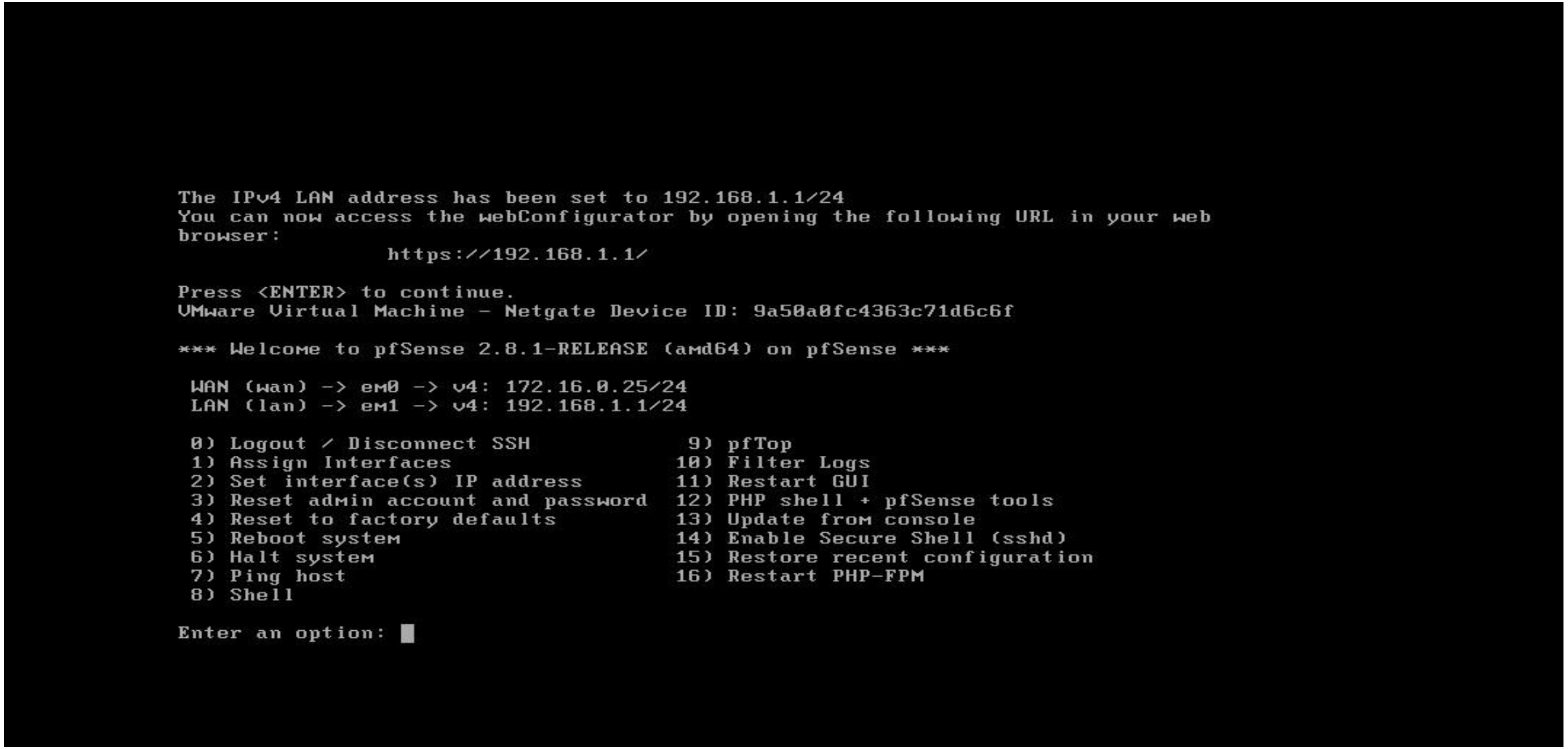
Configure IPv4 address WAN interface via DHCP? (y/n) n

Enter the new WAN IPv4 address.  Press <ENTER> for none:
> 172.16.0.25

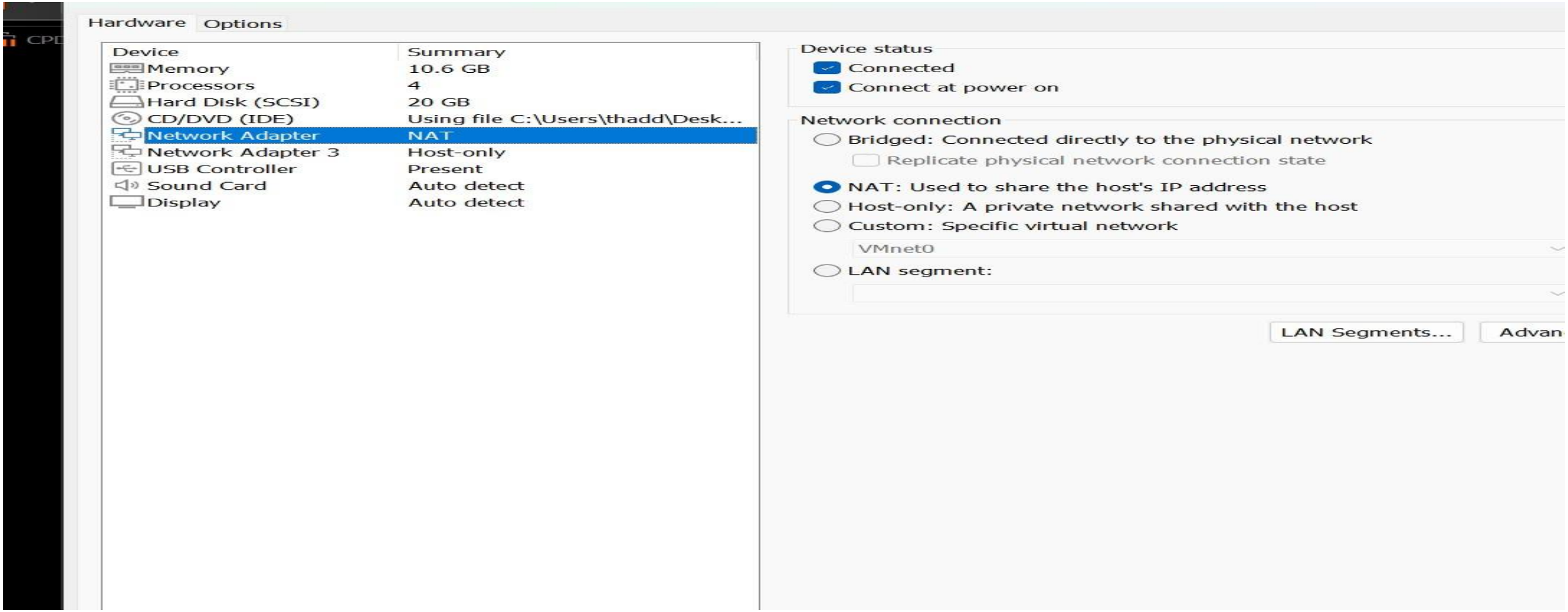
Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new WAN IPv4 subnet bit count (1 to 32):
>
```

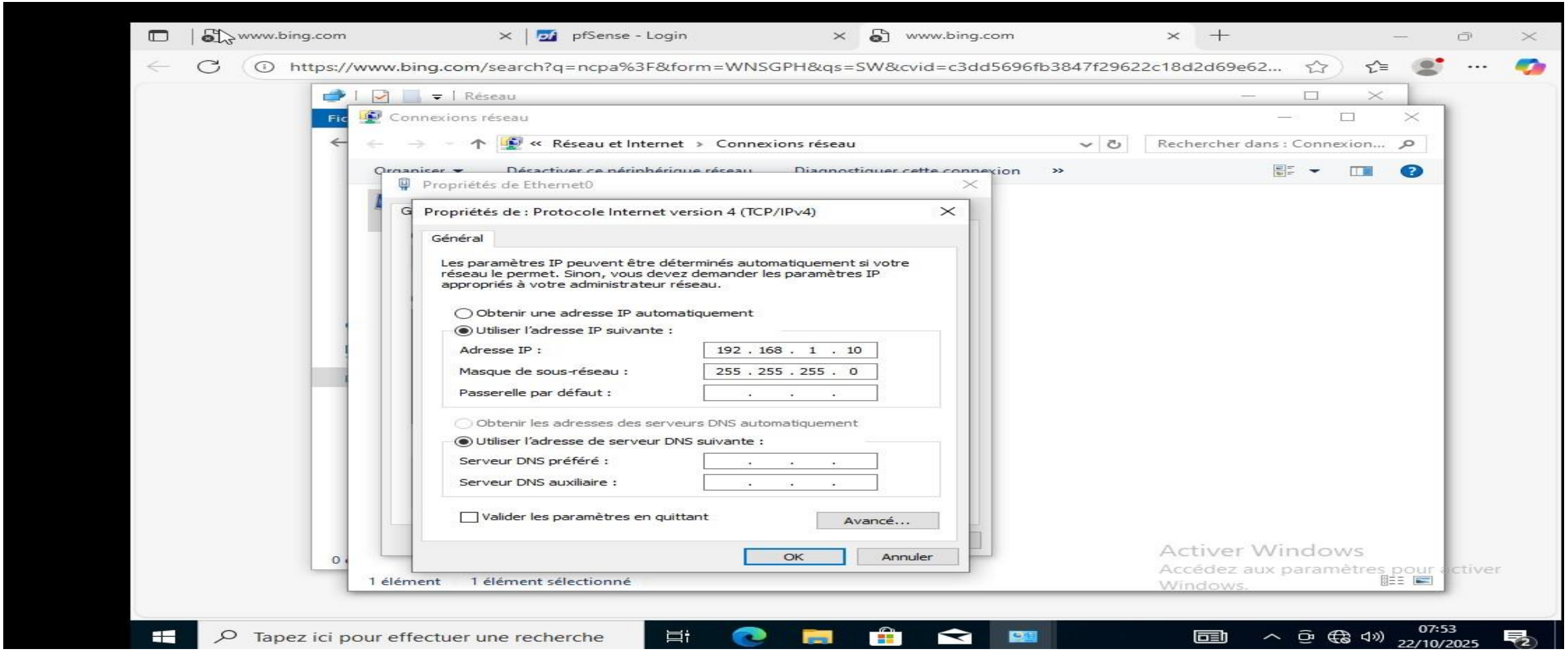
ON DIT NON PAS DE DHCP



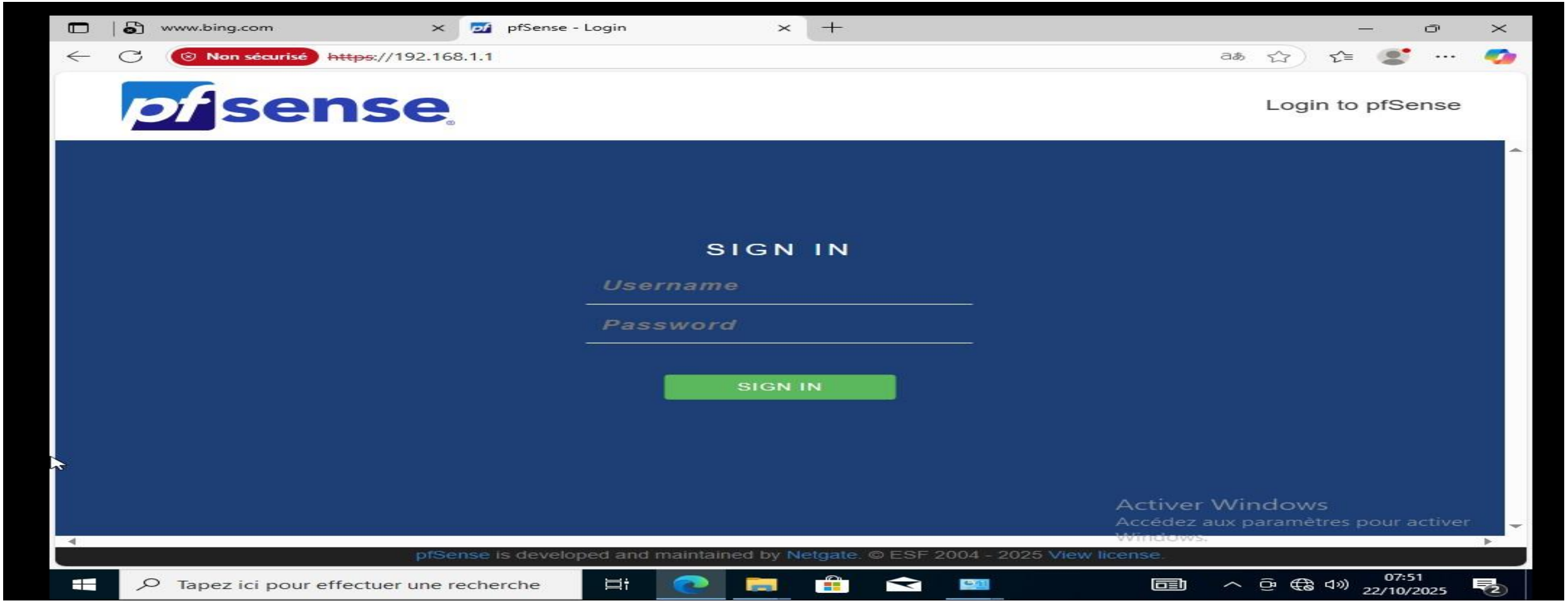
C'EST LA CONFIGURATION DES CARTE RÉSEAU DE LA MACHINE PFSENSE



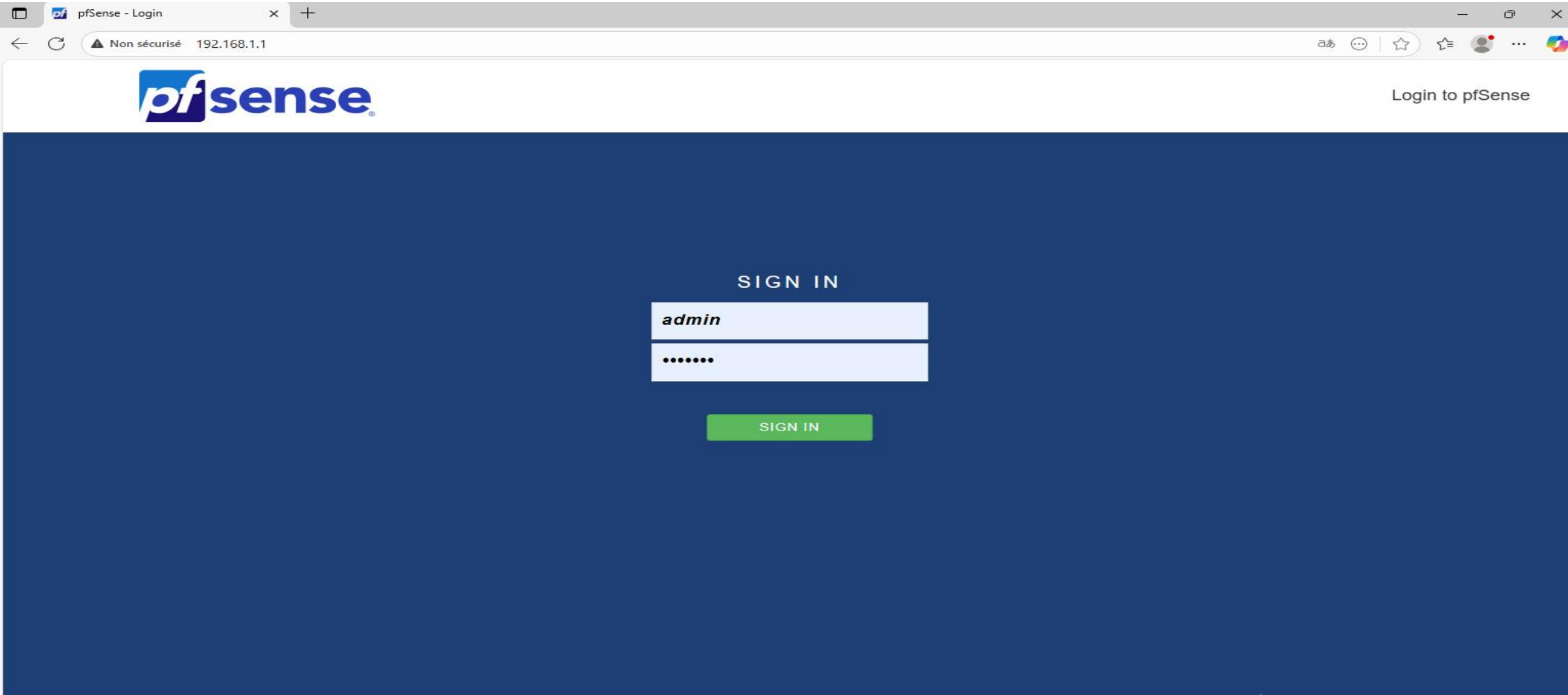
ON METS L'ADRESSE IP DANS LE MÊME RÉSEAU QUE MA MACHINE PFSENSE



J'ACCÈDE À L'INTERFACE WEB GRÂCE AU NAVIGATEUR ET À L'ADRESSE LAN DE PFSENSE



CONFIGURATION DES ADRESSES A L'AIDE DU DHCP



pfSense.home.arpa - Status: Dash

Non sécurisé 192.168.1.1

pfSense COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help 6

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

Status / Dashboard

System Information

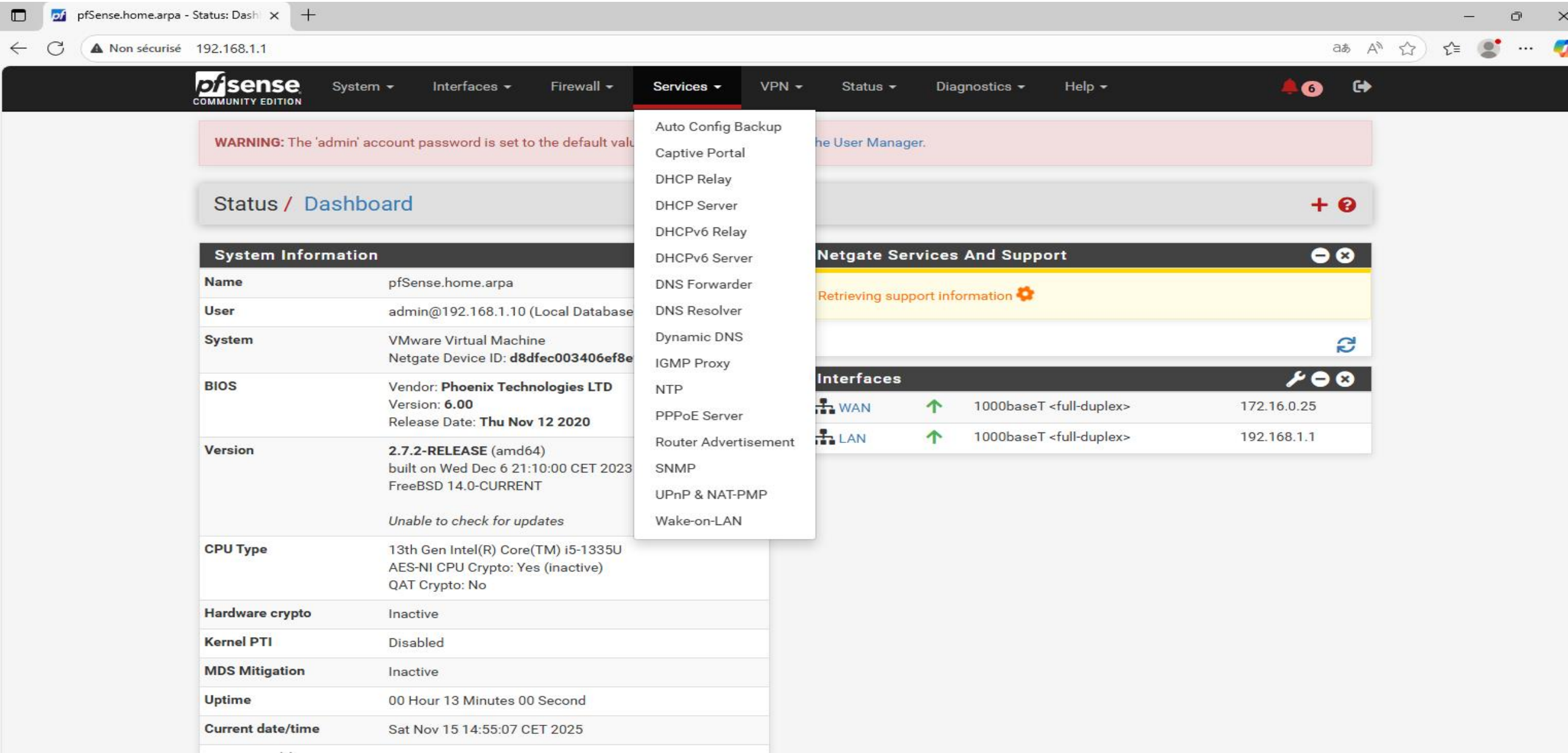
Name	pfSense.home.arpa
User	admin@192.168.1.10 (Local Database)
System	VMware Virtual Machine Netgate Device ID: **d8dfec003406ef8e90aa**
BIOS	Vendor: **Phoenix Technologies LTD** Version: **6.00** Release Date: **Thu Nov 12 2020**
Version	**2.7.2-RELEASE** (amd64) built on Wed Dec 6 21:10:00 CET 2023 FreeBSD 14.0-CURRENT *Unable to check for updates*
CPU Type	13th Gen Intel(R) Core(TM) i5-1335U AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 10 Minutes 39 Seconds
Current date/time	Sat Nov 15 14:52:46 CET 2025

Netgate Services And Support

Retrieving support information

Interfaces

WAN	↑	1000baseT <full-duplex>	172.16.0.25
LAN	↑	1000baseT <full-duplex>	192.168.1.1


pfSense.home.arpa - Status: Dash
Non sécurisé 192.168.1.1
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
6
Auto Config Backup
Captive Portal
DHCP Relay
DHCP Server
DHCPv6 Relay
DHCPv6 Server
DNS Forwarder
DNS Resolver
Dynamic DNS
IGMP Proxy
NTP
PPPoE Server
Router Advertisement
SNMP
UPnP & NAT-PMP
Wake-on-LAN
WARNING: The 'admin' account password is set to the default valu
he User Manager.
Status / Dashboard
System Information
Name pfSense.home.arpa
User admin@192.168.1.10 (Local Database
System VMware Virtual Machine
Netgate Device ID: d8dfec003406ef8e
BIOS Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Thu Nov 12 2020
Version 2.7.2-RELEASE (amd64)
built on Wed Dec 6 21:10:00 CET 2023
FreeBSD 14.0-CURRENT
Unable to check for updates
CPU Type 13th Gen Intel(R) Core(TM) i5-1335U
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto Inactive
Kernel PTI Disabled
MDS Mitigation Inactive
Uptime 00 Hour 13 Minutes 00 Second
Current date/time Sat Nov 15 14:55:07 CET 2025
Netgate Services And Support
Retrieving support information
Interfaces
WAN 1000baseT <full-duplex> 172.16.0.25
LAN 1000baseT <full-duplex> 192.168.1.1

pfSense.home.arpa - Services: DH

Non sécurisé 192.168.1.1/services_dhcp.php

pfSense COMMUNITY EDITION | System | Interfaces | Firewall | Services | VPN | Status | Diagnostics | Help | 6

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

Services / DHCP Server / LAN

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.

WAN | LAN

General DHCP Options

DHCP Backend	ISC DHCP
Enable	☐ Enable DHCP server on LAN interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on ***any*** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

pfSense.home.arpa - Services: DH

Non sécurisé 192.168.1.1/services_dhcp.php

Deny Unknown Clients

Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on ***any*** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

Ignore Denied Clients

☐ Ignore denied clients rather than reject

This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.

Ignore Client Identifiers

☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request

This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet 192.168.1.0/24

Subnet Range 192.168.1.1 - 192.168.1.254

Address Pool Range

192.168.1.10 — From

192.168.1.245 — To

The specified range for this pool must not be within the range configured on any other address pool for this interface.

Additional Pools

+ Add Address Pool

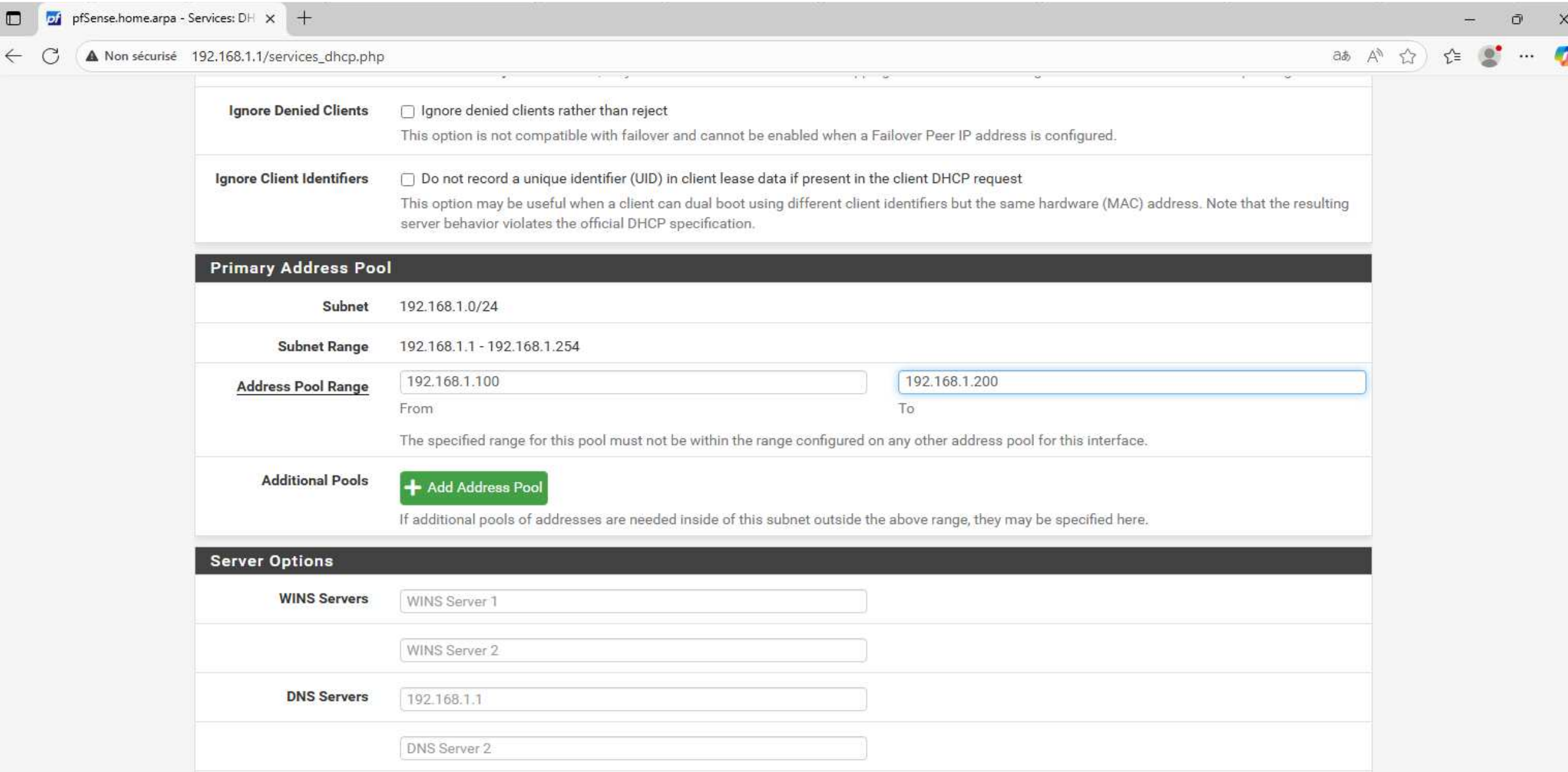
If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.

Server Options

WINS Servers

WINS Server 1

WINS Server 2


pfSense.home.arpa - Services: DH
Non sécurisé
192.168.1.1/services_dhcp.php
Ignore Denied Clients
Ignore denied clients rather than reject
This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers
Do not record a unique identifier (UID) in client lease data if present in the client DHCP request
This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.
Primary Address Pool
Subnet
192.168.1.0/24
Subnet Range
192.168.1.1 - 192.168.1.254
Address Pool Range
192.168.1.100
From
192.168.1.200
To
The specified range for this pool must not be within the range configured on any other address pool for this interface.
Additional Pools
Add Address Pool
If additional pools of addresses are needed inside of this subnet outside the above range, they may be specified here.
Server Options
WINS Servers
WINS Server 1
WINS Server 2
DNS Servers
192.168.1.1
DNS Server 2

Other DHCP Options

Gateway

192.168.1.1

The default is to use the IP address of this firewall interface as the gateway. Specify an alternate gateway here if this is not the correct gateway for the network. Enter "none" for no gateway assignment.

Domain Name

home.arpa

The default is to use the domain name of this firewall as the default domain name provided by DHCP. An alternate domain name may be specified here.

Domain Search List

example.com;sub.example.com

The DHCP server can optionally provide a domain search list. Use the semicolon character as separator.

Default Lease Time

7200

This is used for clients that do not ask for a specific expiration time. The default is 7200 seconds.

Maximum Lease Time

86400

This is the maximum lease time for clients that ask for a specific expiration time. The default is 86400 seconds.

Failover peer IP

Leave blank to disable. Enter the interface IP address of the other firewall (failover peer) in this subnet. Firewalls must be using CARP. Advertising skew of the CARP VIP on this interface determines whether the DHCP daemon is Primary or Secondary. Ensure the advertising skew for the VIP on one firewall is < 20 and the other is > 20.

Static ARP

☐ Enable Static ARP entries

Restricts communication with the firewall to only hosts listed in static mappings containing both IP addresses and MAC addresses. No other hosts will be able to communicate with the firewall on this interface. This behavior is enforced even when DHCP server is disabled.

Time format change

☐ Change DHCP display lease time from UTC to local time

By default DHCP leases are displayed in UTC time. By checking this box DHCP lease time will be displayed in local time and set to the time zone selected. This will be used for all DHCP interfaces lease time.

Statistics graphs

☐ Enable monitoring graphs for DHCP lease statistics

Enable this to add DHCP leases statistics to the Monitoring graphs. Disabled by default.

Activer Windows

pfSense.home.arpa - Services: DH
Non sécurisé 192.168.1.1/services_dhcp.php

By default DHCP leases are displayed in UTC time. By checking this box DHCP lease time will be displayed in local time and set to the time zone selected. This will be used for all DHCP interfaces lease time.

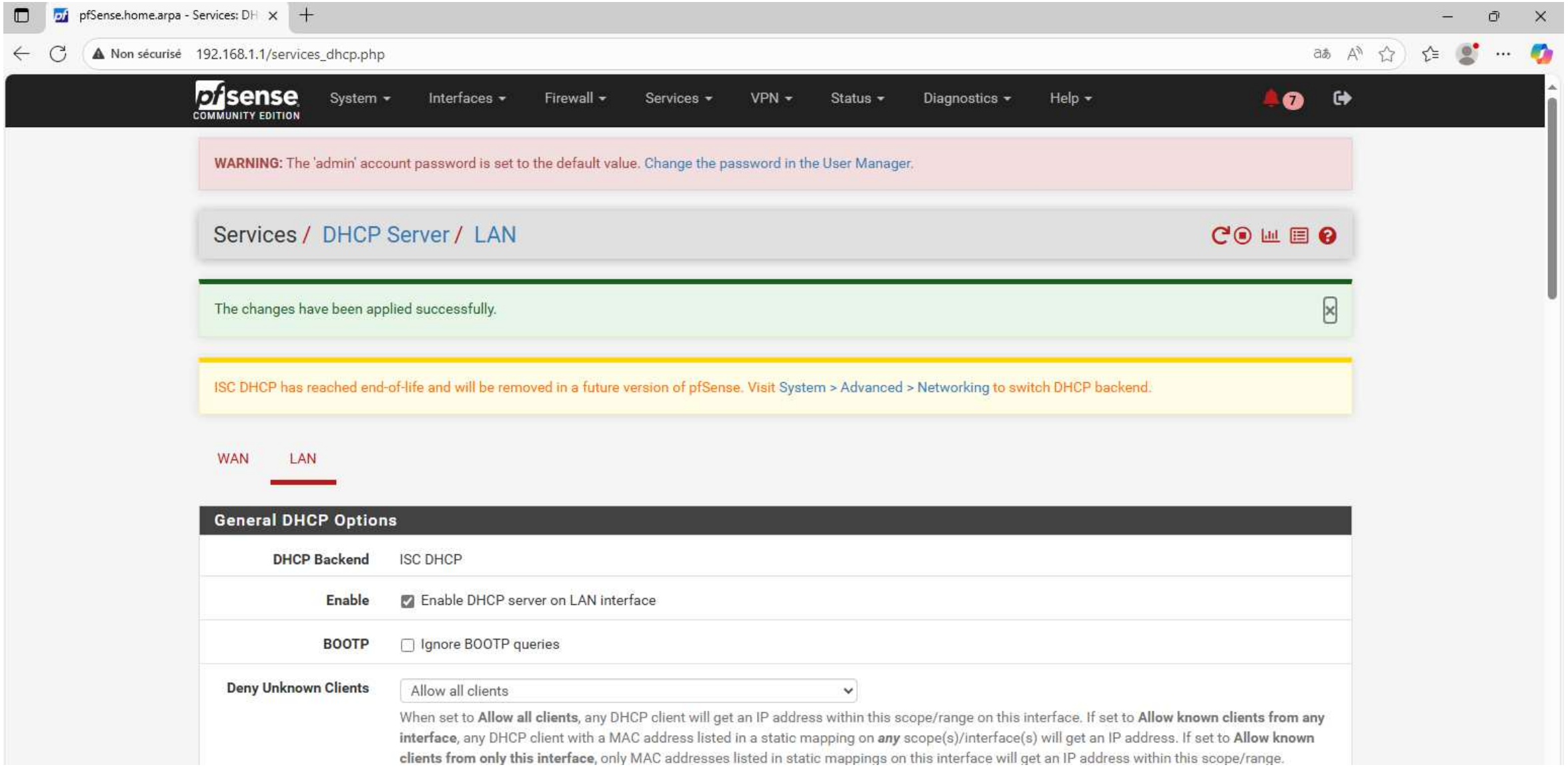
Statistics graphs	☐ Enable monitoring graphs for DHCP lease statistics Enable this to add DHCP leases statistics to the Monitoring graphs. Disabled by default.
Ping check	☐ Disable ping check When enabled dhcpd sends a ping to the address being assigned, and if no response has been heard, it assigns the address. Enabled by default.
Dynamic DNS	Display Advanced
MAC Address Control	Display Advanced
NTP	Display Advanced
TFTP	Display Advanced
LDAP	Display Advanced
Network Booting	Display Advanced
Custom DHCP Options	Display Advanced

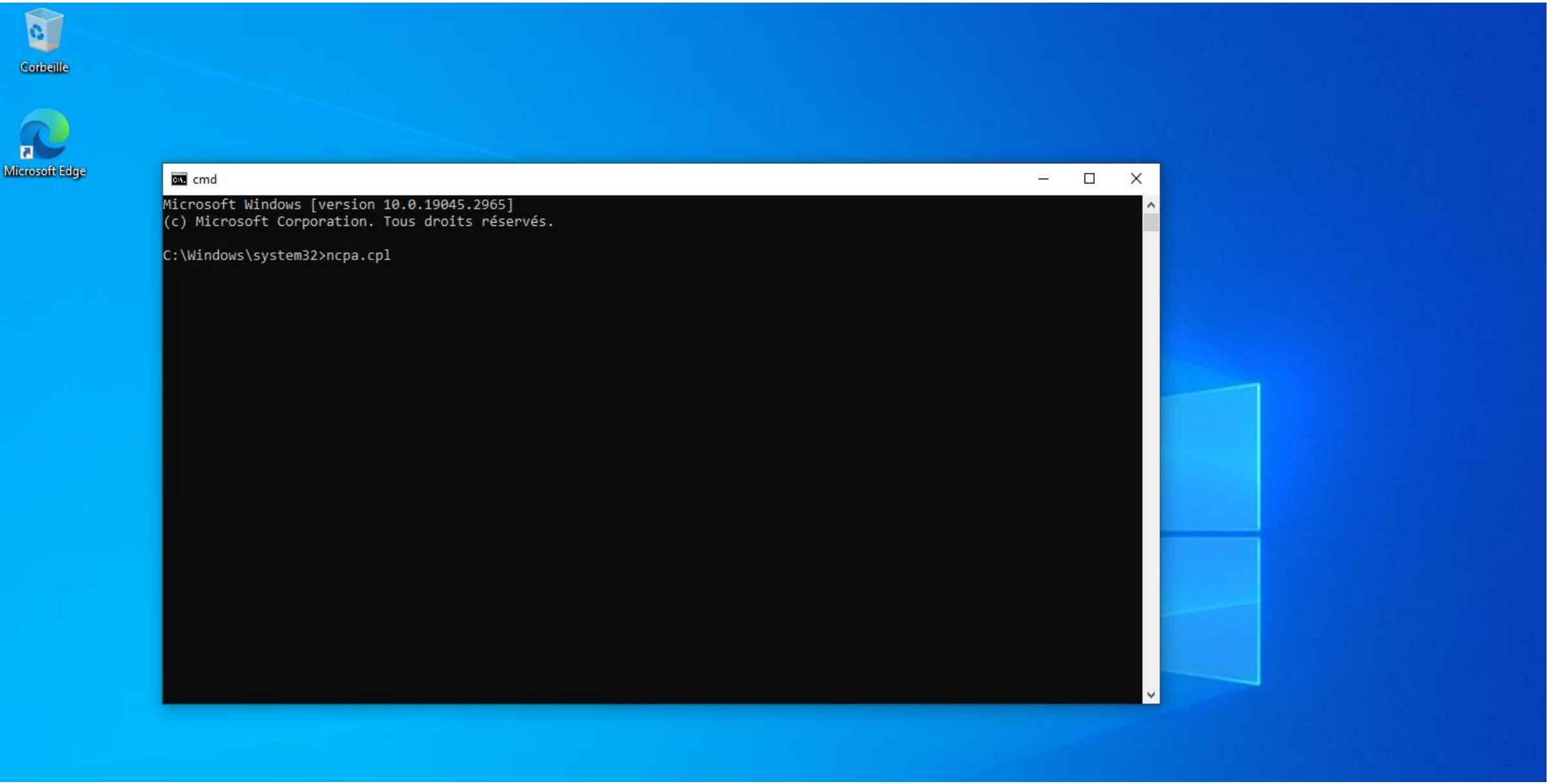
Save

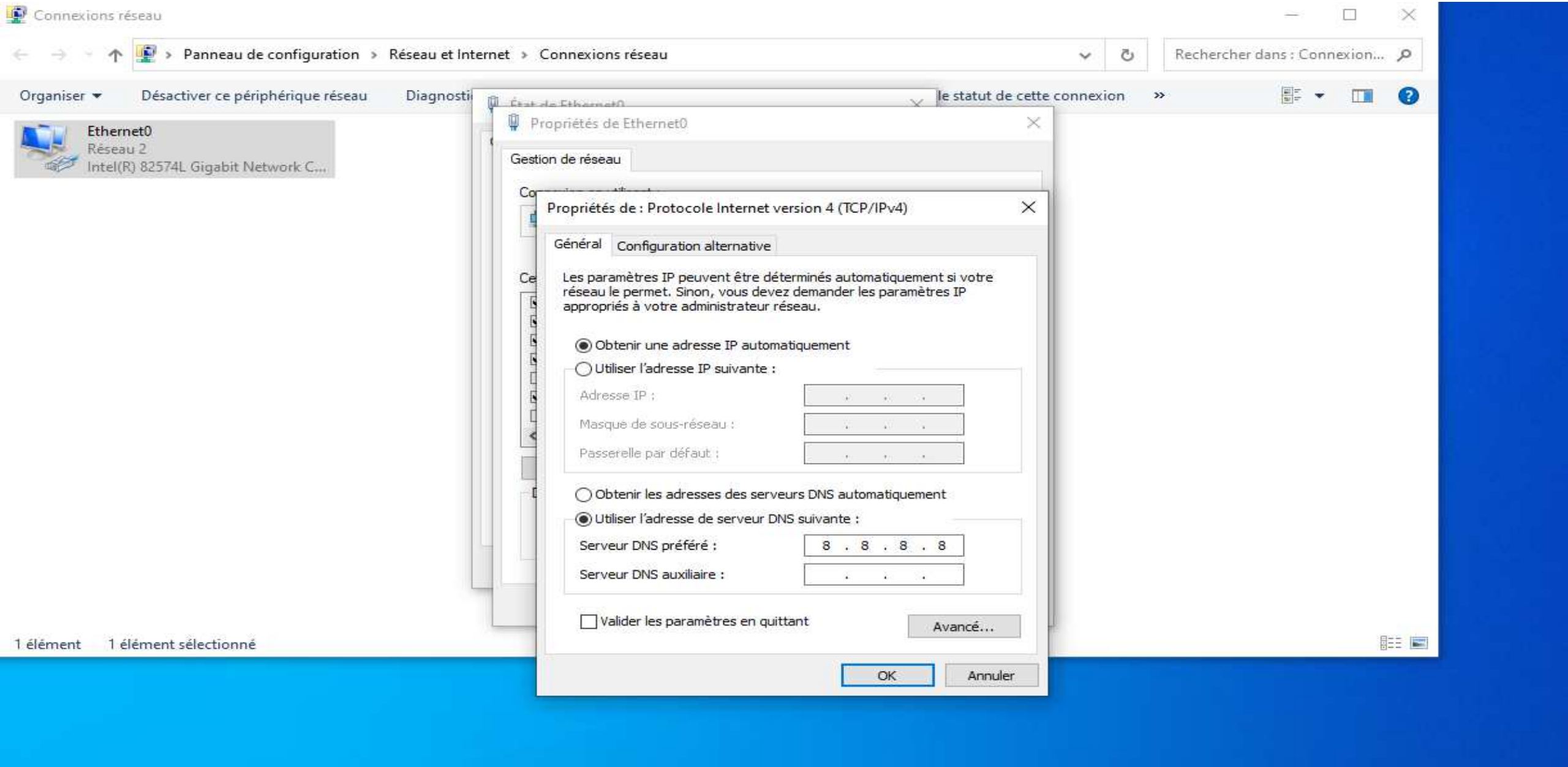
DHCP Static Mappings

Static ARP	MAC address	IP address	Hostname	Description

Add Static Mapping


pfSense.home.arpa - Services: DH
Non sécurisé 192.168.1.1/services_dhcp.php
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
7
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / DHCP Server / LAN
The changes have been applied successfully.
ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.
WAN
LAN
General DHCP Options
DHCP Backend
ISC DHCP
Enable
Enable DHCP server on LAN interface
BOOTP
Ignore BOOTP queries
Deny Unknown Clients
Allow all clients
When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.


Corbeille
Microsoft Edge
cmd
Microsoft Windows [version 10.0.19045.2965]
(c) Microsoft Corporation. Tous droits réservés.
C:\Windows\system32>ncpa.cpl


Connexions réseau
Panneau de configuration › Réseau et Internet › Connexions réseau
Rechercher dans : Connexion...
Organiser
Désactiver ce périphérique réseau
Diagnosti
le statut de cette connexion
Ethernet0
Réseau 2
Intel(R) 82574L Gigabit Network C...
Propriétés de Ethernet0
Gestion de réseau
Propriétés de : Protocole Internet version 4 (TCP/IPv4)
Général
Configuration alternative
Les paramètres IP peuvent être déterminés automatiquement si votre réseau le permet. Sinon, vous devez demander les paramètres IP appropriés à votre administrateur réseau.
Obtenir une adresse IP automatiquement
Utiliser l'adresse IP suivante :
Adresse IP :
Masque de sous-réseau :
Passerelle par défaut :
Obtenir les adresses des serveurs DNS automatiquement
Utiliser l'adresse de serveur DNS suivante :
Serveur DNS préféré : 8 . 8 . 8 . 8
Serveur DNS auxiliaire :
Valider les paramètres en quittant
Avancé...
OK
Annuler
1 élément
1 élément sélectionné

```
Microsoft Windows [version 10.0.19045.2965]
(c) Microsoft Corporation. Tous droits réservés.

C:\Windows\system32>ncpa.cpl

C:\Windows\system32>ipconfig /all

Configuration IP de Windows

   Nom de l'hôte . . . . . . . . . . . . : DESKTOP-EMR6463
   Suffixe DNS principal . . . . . . . . :
   Type de noeud. . . . . . . . . . . . . : Hybride
   Routage IP activé . . . . . . . . . . : Non
   Proxy WINS activé . . . . . . . . . . : Non
   Liste de recherche du suffixe DNS.: home.arpa

Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . : home.arpa
   Description. . . . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Adresse physique . . . . . . . . . . . : 00-0C-29-DF-B3-77
   DHCP activé. . . . . . . . . . . . . . : Oui
   Configuration automatique activée. . . : Oui
```

```
Configuration IP de Windows


Carte Ethernet Ethernet0 :

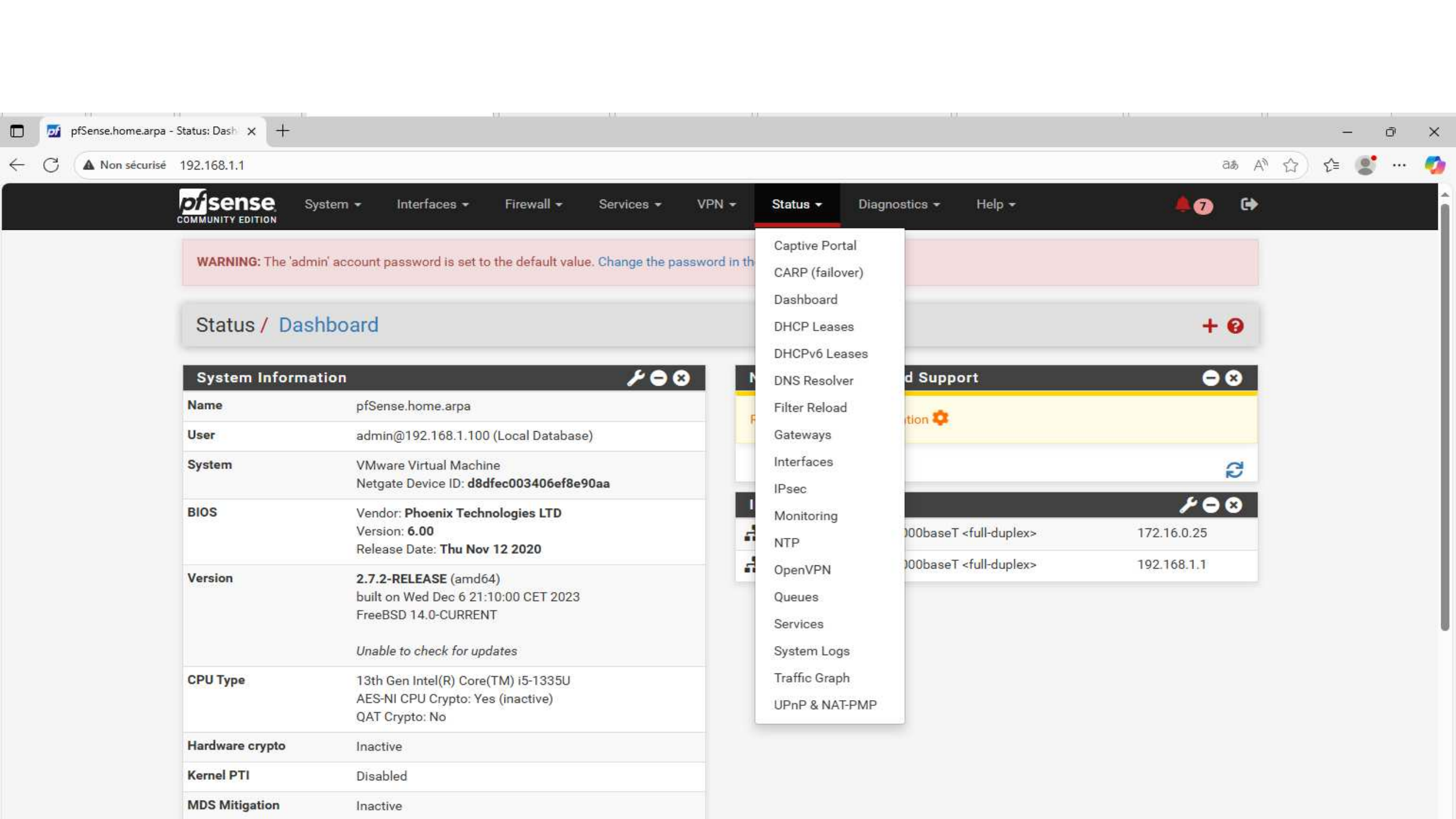
   Suffixe DNS propre à la connexion. . . :
   Passerelle par défaut. . . . . . . . . :

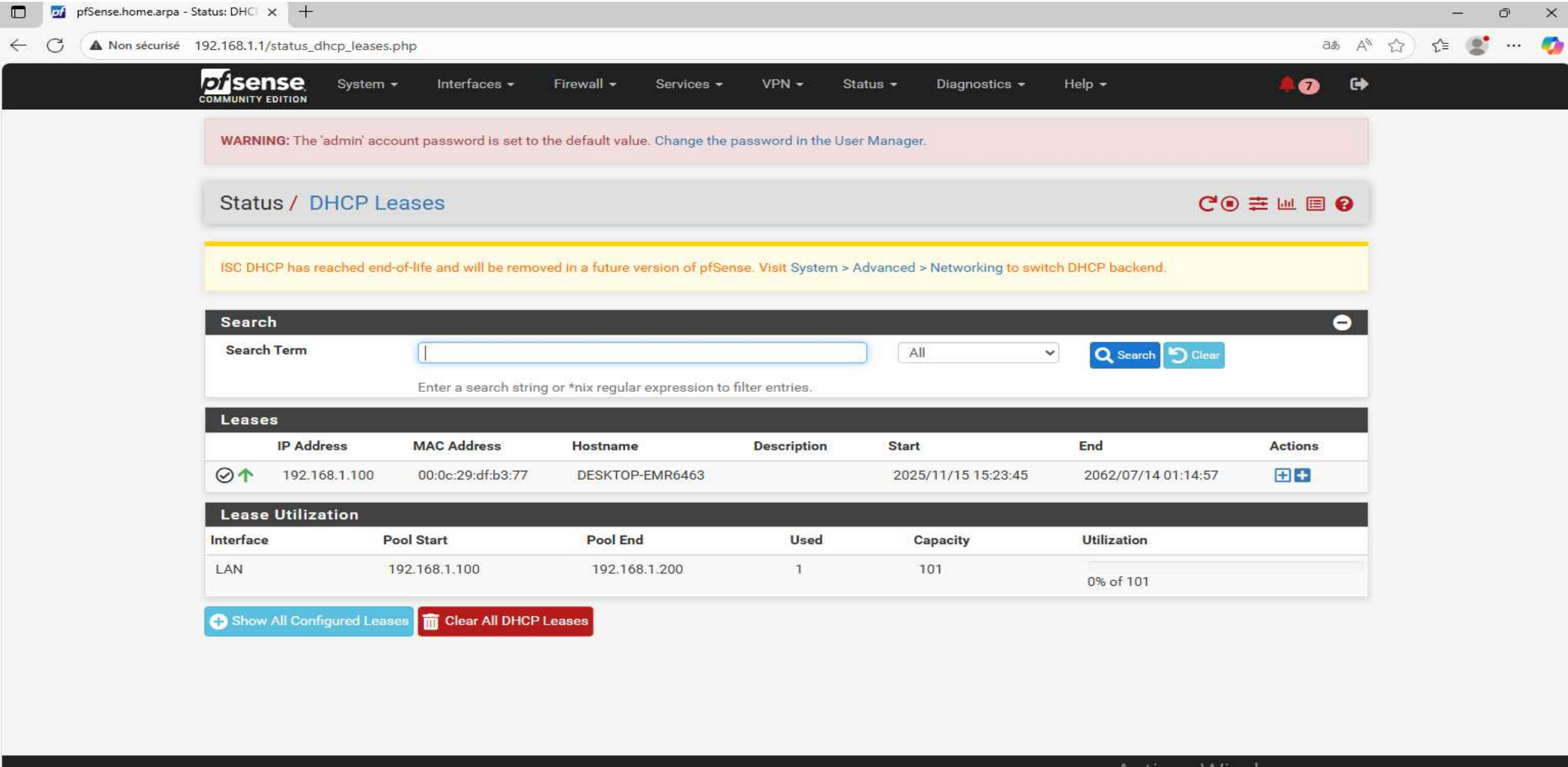
C:\Windows\system32>ipconfig /renew

Configuration IP de Windows


Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . : home.arpa
   Adresse IPv4. . . . . . . . . . . . . .: 192.168.1.100
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . :
```


pfSense.home.arpa - Status: Dash
Non sécurisé 192.168.1.1
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
7
Captive Portal
CARP (failover)
Dashboard
DHCP Leases
DHCPv6 Leases
DNS Resolver
Filter Reload
Gateways
Interfaces
IPsec
Monitoring
NTP
OpenVPN
Queues
Services
System Logs
Traffic Graph
UPnP & NAT-PMP
WARNING: The 'admin' account password is set to the default value. Change the password in th
Status / Dashboard
System Information
Name pfSense.home.arpa
User admin@192.168.1.100 (Local Database)
System VMware Virtual Machine
Netgate Device ID: d8dfec003406ef8e90aa
BIOS Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Thu Nov 12 2020
Version 2.7.2-RELEASE (amd64)
built on Wed Dec 6 21:10:00 CET 2023
FreeBSD 14.0-CURRENT
Unable to check for updates
CPU Type 13th Gen Intel(R) Core(TM) i5-1335U
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto Inactive
Kernel PTI Disabled
MDS Mitigation Inactive
d Support
000baseT <full-duplex> 172.16.0.25
000baseT <full-duplex> 192.168.1.1


pfSense.home.arpa - Status: DHC
Non sécurisé
192.168.1.1/status_dhcp_leases.php
pfsense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
7
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Status / DHCP Leases
ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.
Search
Search Term
All
Search
Clear
Enter a search string or *nix regular expression to filter entries.
Leases
IP Address
MAC Address
Hostname
Description
Start
End
Actions
192.168.1.100
00:0c:29:df:b3:77
DESKTOP-EMR6463
2025/11/15 15:23:45
2062/07/14 01:14:57
Lease Utilization
Interface
Pool Start
Pool End
Used
Capacity
Utilization
LAN
192.168.1.100
192.168.1.200
1
101
0% of 101
Show All Configured Leases
Clear All DHCP Leases

```
Microsoft Windows [version 10.0.19045.3803]
(c) Microsoft Corporation. Tous droits réservés.

C:\Windows\system32>ncpa.cpl

C:\Windows\system32>ipconfig /release

Configuration IP de Windows


Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . :
   Adresse IPv6 de liaison locale. . . . .: fe80::53cf:3275:79dd:2183%6
   Passerelle par défaut. . . . . :

C:\Windows\system32>ipconfig /renew

Configuration IP de Windows


Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . : home.arpa
   Adresse IPv6 de liaison locale. . . . .: fe80::53cf:3275:79dd:2183%6
   Adresse IPv4. . . . . .: 192.168.1.101
   Masque de sous-réseau. . . . . : 255.255.255.0
   Passerelle par défaut. . . . . :

C:\Windows\system32>_
```

pfSense.home.arpa - Status: DHC

Non sécurisé 192.168.1.1/status_dhcp_leases.php

pfsense COMMUNITY EDITION | System | Interfaces | Firewall | Services | VPN | Status | Diagnostics | Help | 7

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

Status / DHCP Leases

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit System > Advanced > Networking to switch DHCP backend.

Search

Search Term | All | Search | Clear

Enter a search string or *nix regular expression to filter entries.

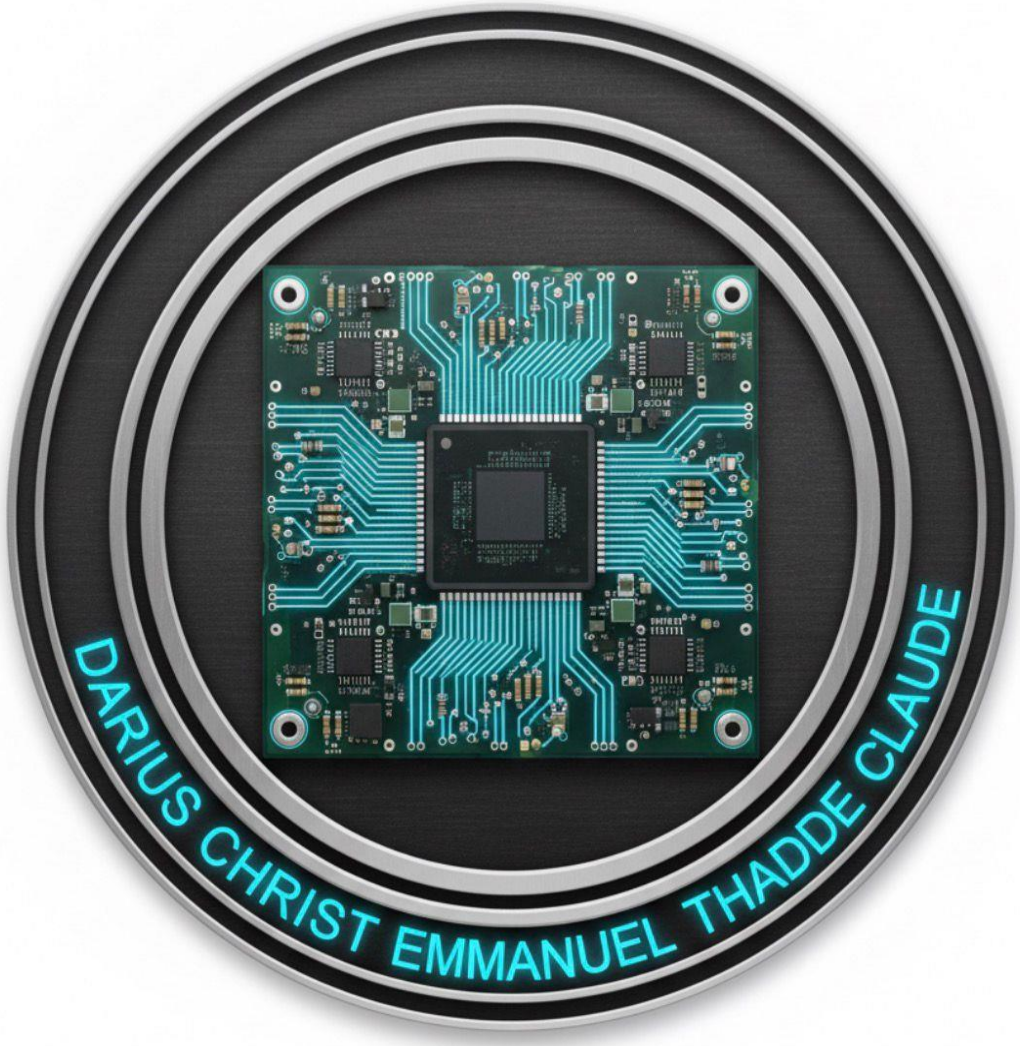
Leases

	IP Address	MAC Address	Hostname	Description	Start	End	Actions
	192.168.1.101	00:0c:29:77:44:42	DESKTOP-ABK06C5		2025/11/15 21:57:25	2062/07/14 07:48:37	
	192.168.1.100	00:0c:29:df:b3:77	DESKTOP-EMR6463		2025/11/15 15:23:45		

Lease Utilization

Interface	Pool Start	Pool End	Used	Capacity	Utilization
LAN	192.168.1.100	192.168.1.200	2	101	1% of 101

Show All Configured Leases | Clear All DHCP Leases



MERCI POUR VOTRE ATTENTION

